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Directory Documentation

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- file camera_component.cpp
- file camera_component.h
- file component.cpp
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- file editor.cpp
- file editor.h
- file engine.cpp
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- file engine_config.h
- file gizmo.h
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- file sound_manager.h
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- file texture.h
- file theme.h
- file transform.cpp
- file transform.h
- file util.cpp
- file util.h

6.2 src Directory Reference

Directories

- directory engine

Chapter 7

Namespace Documentation

7.1 engine Namespace Reference

Namespaces

- namespace editor
- namespace gui

Classes

- class Asset
- class AssetManager
- class CameraComponent
- class Component
- class EditorCameraComponent
- class EngineConfig
- struct EngineCreateInfo
- class EngineInstance
- class InputHandler
- class Material
- class Mesh
- class MeshComponent
- class Model
- class OpenGLRenderer
- class PhysicsHandler
- struct Ray
- class Saveable
- class Scene
- class SceneLoader
- class SchemeScriptEngine
- class ScreenShader
- class ScriptComponent
- class ScriptEngine
- struct ScriptFile
- struct ScriptLiteral
- class Shader
- class Sound

- class SoundManager
- class Texture
- class Transform
- struct Uniform_d
- struct Uniform_f
- struct Uniform_i
- struct Uniform_mat4
- struct Uniform_ui
- struct Uniform_vec3
- class UniformList

Typedefs

- using ConfigValue = std::variant<int, float, std::string,glm::vec2, glm::vec3, glm::vec4>
- typedef std::variant< ScriptLiteral, ScriptFile > ScriptData
- using Uniform

Enumerations

- enum class KeyState { KEY_UP , KEY_PRESSED , KEY_DOWN , KEY_RELEASED }

Functions

- std::string trim_str (std::string in)
- FACTORY_DEF (EditorCameraComponent)
- FACTORY_DEF (CameraComponent)
- FACTORY (EditorCameraComponent)
- FACTORY (CameraComponent)
- EngineInstance * Engine ()
- EngineInstance * CreateEngine (EngineCreateInfo &create_info)
- void InputHandlerKeyCallback (GLFWwindow *window, int key, int scancode, int action, int mods)
- void InputHandlerMousePositionCallback (GLFWwindow *window, double xpos, double ypos)
- void InputHandlerMouseButtonCallback (GLFWwindow *window, int button, int action, int mods)
- int MakeKeyval (int key, int mod)
- FACTORY (MeshComponent)
- Model * GetModel (std::string path)
- void APIENTRY MessageCallback (GLenum source, GLenum type, GLuint id, GLenum severity, GLsizei length, const GLchar *message, const void *userParam)
- void window_size_callback (GLFWwindow *window, int width, int height)
- bool TestRayOBBIntersection (glm::vec3 ray_origin, glm::vec3 ray_direction, glm::vec3 aabb_min, glm::vec3 aabb_max, glm::mat4 ModelMatrix, float &intersection_distance)
- void tick (float dt)
- void init ()

Variables

- EngineInstance * instance = nullptr
- std::map< std::string, Model * > modelCache
- int window_w = 0
- int window_h = 0
- float quad_data [ ]

7.1.1 Typedef Documentation

7.1.1.1 ConfigValue

```
using engine::ConfigValue = std::variant<int, float, std::string,glm::vec2, glm::vec3, glm::vec4>
```

7.1.1.2 ScriptData

```
typedef std::variant<ScriptLiteral, ScriptFile> engine::ScriptData
```

7.1.1.3 Uniform

```
using engine::Uniform
```

Initial value:

```
std::variant<Uniform_vec3,Uniform_i, Uniform_ui, Uniform_f, Uniform_d, Uniform_mat4>
```

7.1.2 Enumeration Type Documentation

7.1.2.1 KeyState

```
enum class engine::KeyState  [strong]
```

Enumerator

KEY_UP	
KEY_PRESSED	
KEY_DOWN	
KEY_RELEASED	

7.1.3 Function Documentation

7.1.3.1 CreateEngine()

```
EngineInstance * engine::CreateEngine (
            EngineCreateInfo & create_info)
```

7.1.3.2 Engine()

```
EngineInstance * engine::Engine ()
```

7.1.3.3 FACTORY() [1/3]

```
engine::FACTORY (
            CameraComponent )
```

7.1.3.4 FACTORY() [2/3]

```
engine::FACTORY (
            EditorCameraComponent )
```

7.1.3.5 FACTORY() [3/3]

```
engine::FACTORY (
            MeshComponent )
```

7.1.3.6 FACTORY_DEF() [1/2]

```
engine::FACTORY_DEF (
            CameraComponent )
```

7.1.3.7 FACTORY_DEF() [2/2]

```
engine::FACTORY_DEF (
            EditorCameraComponent )
```

7.1.3.8 GetModel()

```
Model * engine::GetModel (
            std::string path)
```

7.1.3.9 init()

```
void engine::init ()
```

7.1.3.10 InputHandlerKeyCallback()

```
void engine::InputHandlerKeyCallback (
            GLFWwindow * window,
            int key,
            int scancode,
            int action,
            int mods)
```

7.1.3.11 InputHandlerMouseButtonCallback()

```
void engine::InputHandlerMouseButtonCallback (
            GLFWwindow * window,
            int button,
            int action,
            int mods)
```

7.1.3.12 InputHandlerMousePositionCallback()

```
void engine::InputHandlerMousePositionCallback (
            GLFWwindow * window,
            double xpos,
            double ypos)
```

7.1.3.13 MakeKeyval()

```
int engine::MakeKeyval (
            int key,
            int mod)
```

7.1.3.14 MessageCallback()

```
void APIENTRY engine::MessageCallback (
            GLenum source,
            GLenum type,
            GLuint id,
            GLenum severity,
            GLsizei length,
            const GLchar * message,
            const void * userParam)
```

7.1.3.15 TestRayOBBIntersection()

```
bool engine::TestRayOBBIntersection (
            glm::vec3 ray_origin,
            glm::vec3 ray_direction,
            glm::vec3 aabb_min,
            glm::vec3 aabb_max,
            glm::mat4 ModelMatrix,
            float & intersection_distance)
```

7.1.3.16 tick()

```
void engine::tick (
            float dt)
```

7.1.3.17 trim_str()

```
std::string engine::trim_str (
            std::string in)
```

7.1.3.18 window_size_callback()

```
void engine::window_size_callback (
            GLFWwindow * window,
            int width,
            int height)
```

7.1.4 Variable Documentation

7.1.4.1 instance

```
EngineInstance* engine::instance = nullptr
```

7.1.4.2 modelCache

```
std::map<std::string, Model*> engine::modelCache
```

7.1.4.3 quad_data

```
float engine::quad_data[]
```

Initial value:

```
                = {
    -1.0f, -1.0f, 0.0f,
    -1.0f, 1.0f, 0.0f,
    1.0f, 1.0f, 0.0f,

    -1.0f, -1.0f, 0.0f,
    1.0f, 1.0f, 0.0f,
    1.0f, -1.0f, 0.0f
}
```

7.1.4.4 window_h

```
int engine::window_h = 0
```

7.1.4.5 window_w

```
int engine::window_w = 0
```

7.2 engine::editor Namespace Reference

Classes

- class EditorInstance
- class Gizmo

Enumerations

- enum class PrefsPage { APPEARANCE , GENERAL }

Functions

- ImGui::FileBrowser saveBrowser (ImGuiFileBrowserFlags_EnterNewFilename)

Variables

- bool debug_panel_vis = false
- bool asset_panel_vis = false
- bool hierarchy_vis = false
- bool inspector_vis = false
- bool actions_vis = false
- bool preferences_vis = false
- bool scene_load_menu_open = false
- ImGui::FileBrowser loadBrowser
- int fps_smooth_samples = 100
- PrefsPage current_page = PrefsPage::GENERAL

7.2.1 Enumeration Type Documentation

7.2.1.1 PrefsPage

```
enum class engine::editor::PrefsPage  [strong]
```

Enumerator

APPEARANCE	
GENERAL	

7.2.2 Function Documentation

7.2.2.1 saveBrowser()

```
ImGui::FileBrowser engine::editor::saveBrowser (
            ImGuiFileBrowserFlags_EnterNewFilename )
```

7.2.3 Variable Documentation

7.2.3.1 actions_vis

```
bool engine::editor::actions_vis = false
```

7.2.3.2 asset_panel_vis

```
bool engine::editor::asset_panel_vis = false
```

7.2.3.3 current_page

```
PrefsPage engine::editor::current_page = PrefsPage::GENERAL
```

7.2.3.4 debug_panel_vis

```
bool engine::editor::debug_panel_vis = false
```

7.2.3.5 fps_smooth_samples

```
int engine::editor::fps_smooth_samples = 100
```

7.2.3.6 hierarchy_vis

```
bool engine::editor::hierarchy_vis = false
```

7.2.3.7 inspector_vis

```
bool engine::editor::inspector_vis = false
```

7.2.3.8 loadBrowser

```
ImGui::FileBrowser engine::editor::loadBrowser
```

7.2.3.9 preferences_vis

```
bool engine::editor::preferences_vis = false
```

7.2.3.10 scene_load_menu_open

```
bool engine::editor::scene_load_menu_open = false
```

7.3 engine::gui Namespace Reference

Functions

- void TextCentered (const char *text)

7.3.1 Function Documentation

7.3.1.1 TextCentered()

```
void engine::gui::TextCentered (
            const char * text)
```

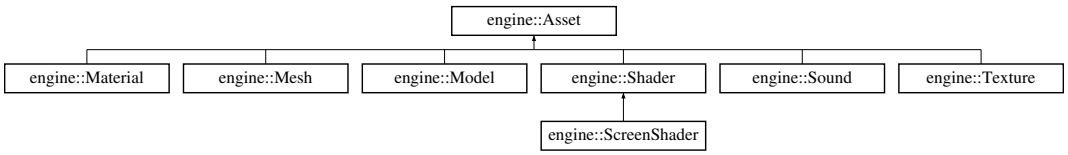
Chapter 8

Class Documentation

8.1 engine::Asset Class Reference

`#include <asset.h>`

Inheritance diagram for engine::Asset:



Public Member Functions

- virtual bool IsValid ()
- std::string GetName ()
- void SetName (std::string name)
- std::string GetPath ()
- void SetPath (std::string path)

Protected Attributes

- bool valid = false
- std::string name = "?????"
- std::string path = "INVALID"

8.1.1 Member Function Documentation

8.1.1.1 GetName()

`std::string engine::Asset::GetName ()`

8.1.1.2 GetPath()

```
std::string engine::Asset::GetPath ()
```

8.1.1.3 IsValid()

```
bool engine::Asset::IsValid ()  [virtual]
```

8.1.1.4 SetName()

```
void engine::Asset::SetName (
            std::string name)
```

8.1.1.5 SetPath()

```
void engine::Asset::SetPath (
            std::string path)
```

8.1.2 Member Data Documentation

8.1.2.1 name

```
std::string engine::Asset::name = "?????"  [protected]
```

8.1.2.2 path

```
std::string engine::Asset::path = "INVALID"  [protected]
```

8.1.2.3 valid

```
bool engine::Asset::valid = false  [protected]
```

The documentation for this class was generated from the following files:

- src/engine/asset.h
- src/engine/asset.cpp

8.2 engine::AssetManager Class Reference

```
#include <asset_manager.h>
```

Public Member Functions

- void Rescan ()
- Mesh * GetMeshOrNull (std::string key)
- Material * GetMaterialOrNull (std::string key)
- Sound * GetSoundOrNull (std::string key)
- Shader * GetShaderOrNull (std::string key)
- Texture * GetTextureOrNull (std::string key)

Public Attributes

- std::map< std::string, Model * > loaded_models
- std::map< std::string, Mesh * > loaded_meshes
- std::map< std::string, Material * > loaded_materials
- std::map< std::string, Texture * > loaded_textures
- std::map< std::string, Shader * > loaded_shaders
- std::map< std::string, Sound * > loaded_sounds

8.2.1 Member Function Documentation

8.2.1.1 GetMaterialOrNull()

```
Material * engine::AssetManager::GetMaterialOrNull (
            std::string key)
```

8.2.1.2 GetMeshOrNull()

```
Mesh * engine::AssetManager::GetMeshOrNull (
            std::string key)
```

8.2.1.3 GetShaderOrNull()

```
Shader * engine::AssetManager::GetShaderOrNull (
            std::string key)
```

8.2.1.4 GetSoundOrNull()

```
Sound * engine::AssetManager::GetSoundOrNull (
            std::string key)
```

8.2.1.5 GetTextureOrNull()

```
Texture * engine::AssetManager::GetTextureOrNull (
            std::string key)
```

8.2.1.6 Rescan()

```
void engine::AssetManager::Rescan ()
```

8.2.2 Member Data Documentation

8.2.2.1 loaded_materials

```
std::map<std::string, Material*> engine::AssetManager::loaded_materials
```

8.2.2.2 loaded_meshes

```
std::map<std::string, Mesh*> engine::AssetManager::loaded_meshes
```

8.2.2.3 loaded_models

```
std::map<std::string, Model*> engine::AssetManager::loaded_models
```

8.2.2.4 loaded_shaders

```
std::map<std::string, Shader *> engine::AssetManager::loaded_shaders
```

8.2.2.5 loaded_sounds

```
std::map<std::string, Sound*> engine::AssetManager::loaded_sounds
```

8.2.2.6 loaded_textures

```
std::map<std::string, Texture*> engine::AssetManager::loaded_textures
```

The documentation for this class was generated from the following files:

- src/engine/asset_manager.h
- src/engine/asset_manager.cpp

8.3 BaseUniforms Struct Reference

```
#include <gl_util.h>
```

Public Attributes

- glm::mat4 model
- glm::mat4 camera

8.3.1 Member Data Documentation

8.3.1.1 camera

```
glm::mat4 BaseUniforms::camera
```

8.3.1.2 model

```
glm::mat4 BaseUniforms::model
```

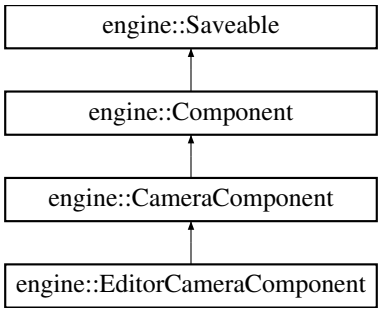
The documentation for this struct was generated from the following file:

- src/engine/gl_util.h

8.4 engine::CameraComponent Class Reference

```
#include <camera_component.h>
```

Inheritance diagram for engine::CameraComponent:



Public Member Functions

- CameraComponent ()
- CameraComponent (std::string name)
- virtual ~CameraComponent ()
- virtual void tick (float dt) override
- virtual glm::mat4 GetCameraMatrix ()
- virtual Ray ScreenPointToRay (glm::vec2 screen_point)
- virtual json Save () override
- virtual bool Load (json value) override
- virtual void DrawWidget () override

Public Member Functions inherited from engine::Component

- Component ()
- Component (std::string name)
- virtual ~Component ()
- virtual void begin_play ()
- virtual void end_play ()
- virtual void render ()
- virtual void editor_render ()
- virtual void init ()
- virtual void destroy ()
- Transform * GetOwner ()
- void SetOwner (Transform *own)
- std::string GetName ()
- uint64_t GetUID ()
- void ComponentDisplay ()

Additional Inherited Members

Public Attributes inherited from engine::Component

- std::string name
- std::string component_type
- uint64_t uid
- Transform * owner

8.4.1 Constructor & Destructor Documentation

8.4.1.1 CameraComponent() [1/2]

```
engine::CameraComponent::CameraComponent ()
```

8.4.1.2 CameraComponent() [2/2]

```
engine::CameraComponent::CameraComponent (
            std::string name)
```

8.4.1.3 ~CameraComponent()

```
engine::CameraComponent::~CameraComponent ()  [virtual]
```

8.4.2 Member Function Documentation

8.4.2.1 DrawWidget()

```
void engine::CameraComponent::DrawWidget ()  [override], [virtual]
```

Implements engine::Component.

Reimplemented in engine::EditorCameraComponent.

8.4.2.2 GetCameraMatrix()

```
glm::mat4 engine::CameraComponent::GetCameraMatrix ()  [virtual]
```

8.4.2.3 Load()

```
bool engine::CameraComponent::Load (
            json value)  [override], [virtual]
```

Implements engine::Component.

Reimplemented in engine::EditorCameraComponent.

8.4.2.4 Save()

```
json engine::CameraComponent::Save ()  [override], [virtual]
```

Implements engine::Component.

Reimplemented in engine::EditorCameraComponent.

8.4.2.5 ScreenPointToRay()

```
Ray engine::CameraComponent::ScreenPointToRay (
            glm::vec2 screen_point)  [virtual]
```

8.4.2.6 tick()

```
void engine::CameraComponent::tick (
            float dt)  [override], [virtual]
```

Reimplemented from engine::Component.

Reimplemented in engine::EditorCameraComponent.

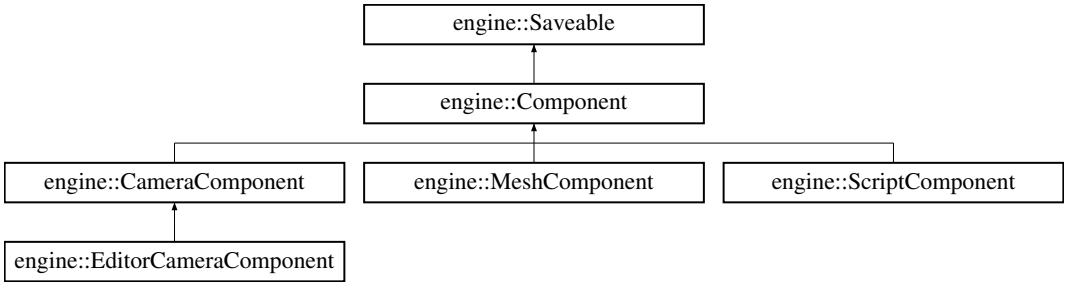
The documentation for this class was generated from the following files:

- src/engine/camera_component.h
- src/engine/camera_component.cpp

8.5 engine::Component Class Reference

```
#include <component.h>
```

Inheritance diagram for engine::Component:



Public Member Functions

- Component ()
- Component (std::string name)
- virtual ~Component ()
- virtual void begin_play ()
- virtual void end_play ()
- virtual void render ()
- virtual void editor_render ()
- virtual void tick (float dt)
- virtual void init ()
- virtual void destroy ()
- virtual void DrawWidget ()=0
- Transform * GetOwner ()
- void SetOwner (Transform *own)
- std::string GetName ()
- virtual json Save () override=0
- virtual bool Load (json value) override=0
- uint64_t GetUID ()
- void ComponentDisplay ()

Public Attributes

- std::string name
- std::string component_type
- uint64_t uid
- Transform * owner

8.5.1 Constructor & Destructor Documentation

8.5.1.1 Component() [1/2]

```
engine::Component::Component ()
```

8.5.1.2 Component() [2/2]

```
engine::Component::Component (
            std::string name)
```

8.5.1.3 ~Component()

```
engine::Component::~Component ()  [virtual]
```

8.5.2 Member Function Documentation

8.5.2.1 begin_play()

```
void engine::Component::begin_play ()  [virtual]
```

8.5.2.2 ComponentDisplay()

```
void engine::Component::ComponentDisplay ()
```

8.5.2.3 destroy()

```
void engine::Component::destroy ()  [virtual]
```

Reimplemented in engine::MeshComponent, and engine::ScriptComponent.

8.5.2.4 DrawWidget()

```
virtual void engine::Component::DrawWidget ()  [pure virtual]
```

Implemented in engine::CameraComponent, engine::EditorCameraComponent, and engine::MeshComponent.

8.5.2.5 editor_render()

```
void engine::Component::editor_render ()  [virtual]
```

8.5.2.6 end_play()

```
void engine::Component::end_play ()  [virtual]
```

8.5.2.7 GetName()

```
std::string engine::Component::GetName ()
```

8.5.2.8 GetOwner()

```
Transform * engine::Component::GetOwner ()
```

8.5.2.9 GetUID()

```
uint64_t engine::Component::GetUID ()
```

8.5.2.10 init()

```
void engine::Component::init ()  [virtual]
```

Reimplemented in engine::MeshComponent, and engine::ScriptComponent.

8.5.2.11 Load()

```
virtual bool engine::Component::Load (
            json value)  [override], [pure virtual]
```

Implements engine::Saveable.

Implemented in engine::CameraComponent, engine::EditorCameraComponent, and engine::MeshComponent.

8.5.2.12 render()

```
void engine::Component::render ()  [virtual]
```

Reimplemented in engine::MeshComponent.

8.5.2.13 Save()

```
json engine::Component::Save ()  [override], [pure virtual]
```

Implements engine::Saveable.

Implemented in engine::CameraComponent, engine::EditorCameraComponent, and engine::MeshComponent.

8.5.2.14 SetOwner()

```
void engine::Component::SetOwner (
            Transform * own)
```

8.5.2.15 tick()

```
void engine::Component::tick (
            float dt)  [virtual]
```

Reimplemented in engine::CameraComponent, engine::EditorCameraComponent, and engine::ScriptComponent.

8.5.3 Member Data Documentation

8.5.3.1 component_type

```
std::string engine::Component::component_type
```

8.5.3.2 name

```
std::string engine::Component::name
```

8.5.3.3 owner

```
Transform* engine::Component::owner
```

8.5.3.4 uid

```
uint64_t engine::Component::uid
```

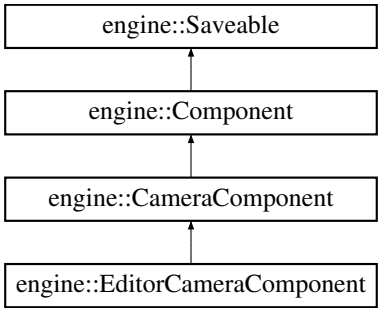
The documentation for this class was generated from the following files:

- src/engine/component.h
- src/engine/component.cpp

8.6 engine::EditorCameraComponent Class Reference

```
#include <camera_component.h>
```

Inheritance diagram for engine::EditorCameraComponent:



Public Member Functions

- EditorCameraComponent ()
- EditorCameraComponent (std::string name)
- virtual ~EditorCameraComponent ()
- virtual void tick (float dt) override
- virtual void DrawWidget () override
- virtual json Save () override
- virtual bool Load (json value) override

Public Member Functions inherited from engine::CameraComponent

- CameraComponent ()
- CameraComponent (std::string name)
- virtual ~CameraComponent ()
- virtual glm::mat4 GetCameraMatrix ()
- virtual Ray ScreenPointToRay (glm::vec2 screen_point)

Public Member Functions inherited from engine::Component

- Component ()
- Component (std::string name)
- virtual ~Component ()
- virtual void begin_play ()
- virtual void end_play ()
- virtual void render ()
- virtual void editor_render ()
- virtual void init ()
- virtual void destroy ()
- Transform * GetOwner ()
- void SetOwner (Transform *own)
- std::string GetName ()
- uint64_t GetUID ()
- void ComponentDisplay ()

Protected Attributes

- float movement_speed = 25.0f

Additional Inherited Members

Public Attributes inherited from engine::Component

- std::string name
- std::string component_type
- uint64_t uid
- Transform * owner

8.6.1 Constructor & Destructor Documentation

8.6.1.1 EditorCameraComponent() [1/2]

```
engine::EditorCameraComponent::EditorCameraComponent ()
```

8.6.1.2 EditorCameraComponent() [2/2]

```
engine::EditorCameraComponent::EditorCameraComponent (
            std::string name)
```

8.6.1.3 ~EditorCameraComponent()

```
engine::EditorCameraComponent::~EditorCameraComponent ()  [virtual]
```

8.6.2 Member Function Documentation

8.6.2.1 DrawWidget()

```
void engine::EditorCameraComponent::DrawWidget ()  [override], [virtual]
```

Reimplemented from engine::CameraComponent.

8.6.2.2 Load()

```
bool engine::EditorCameraComponent::Load (
            json value)  [override], [virtual]
```

Reimplemented from engine::CameraComponent.

8.6.2.3 Save()

```
json engine::EditorCameraComponent::Save ()  [override], [virtual]
```

Reimplemented from engine::CameraComponent.

8.6.2.4 tick()

```
void engine::EditorCameraComponent::tick (
            float dt)  [override], [virtual]
```

Reimplemented from engine::CameraComponent.

8.6.3 Member Data Documentation

8.6.3.1 movement_speed

```
float engine::EditorCameraComponent::movement_speed = 25.0f  [protected]
```

The documentation for this class was generated from the following files:

- src/engine/camera_component.h
- src/engine/camera_component.cpp

8.7 engine::editor::EditorInstance Class Reference

```
#include <editor.h>
```

Public Member Functions

- EditorInstance ()
- ~EditorInstance ()
- void Run ()
- void DebugPanel ()
- void ActionsPanel ()
- void AssetPanel ()
- void Inspector ()
- void SceneHierarchy ()
- void MenuBar ()
- void TransportPanel ()
- void OpenSceneLoadMenu ()
- void OpenSceneSaveMenu ()
- void PreferencesWindow ()
- void DrawTransformGizmo ()
- void SceneLoadMenu ()
- void DrawGrid ()

Public Attributes

- EngineInstance * engine

8.7.1 Constructor & Destructor Documentation

8.7.1.1 EditorInstance()

```
engine::editor::EditorInstance::EditorInstance ()
```

8.7.1.2 ~EditorInstance()

```
engine::editor::EditorInstance::~EditorInstance ()
```

8.7.2 Member Function Documentation

8.7.2.1 ActionsPanel()

```
void engine::editor::EditorInstance::ActionsPanel ()
```

8.7.2.2 AssetPanel()

```
void engine::editor::EditorInstance::AssetPanel ()
```

8.7.2.3 DebugPanel()

```
void engine::editor::EditorInstance::DebugPanel ()
```

8.7.2.4 DrawGrid()

```
void engine::editor::EditorInstance::DrawGrid ()
```

8.7.2.5 DrawTransformGizmo()

```
void engine::editor::EditorInstance::DrawTransformGizmo ()
```

8.7.2.6 Inspector()

```
void engine::editor::EditorInstance::Inspector ()
```

8.7.2.7 MenuBar()

```
void engine::editor::EditorInstance::MenuBar ()
```

8.7.2.8 OpenSceneLoadMenu()

```
void engine::editor::EditorInstance::OpenSceneLoadMenu ()
```

8.7.2.9 OpenSceneSaveMenu()

```
void engine::editor::EditorInstance::OpenSceneSaveMenu ()
```

8.7.2.10 PreferencesWindow()

```
void engine::editor::EditorInstance::PreferencesWindow ()
```

8.7.2.11 Run()

```
void engine::editor::EditorInstance::Run ()
```

8.7.2.12 SceneHierarchy()

```
void engine::editor::EditorInstance::SceneHierarchy ()
```

8.7.2.13 SceneLoadMenu()

```
void engine::editor::EditorInstance::SceneLoadMenu ()
```

8.7.2.14 TransportPanel()

```
void engine::editor::EditorInstance::TransportPanel ()
```

8.7.3 Member Data Documentation

8.7.3.1 engine

```
EngineInstance* engine::editor::EditorInstance::engine
```

The documentation for this class was generated from the following files:

- src/engine/editor.h
- src/engine/editor.cpp

8.8 engine::EngineConfig Class Reference

```
#include <engine_config.h>
```

Public Member Functions

- EngineConfig ()
- EngineConfig (json value)
- ~EngineConfig ()
- ConfigValue GetConfigValue (std::string val)

Public Attributes

- std::map< std::string, ConfigValue > current_config

8.8.1 Constructor & Destructor Documentation

8.8.1.1 EngineConfig() [1/2]

```
engine::EngineConfig::EngineConfig ()
```

8.8.1.2 EngineConfig() [2/2]

```
engine::EngineConfig::EngineConfig (
            json value)
```

8.8.1.3 ~EngineConfig()

```
engine::EngineConfig::~EngineConfig ()
```

8.8.2 Member Function Documentation

8.8.2.1 GetConfigValue()

```
ConfigValue engine::EngineConfig::GetConfigValue (
            std::string val)
```

8.8.3 Member Data Documentation

8.8.3.1 current_config

```
std::map<std::string,ConfigValue> engine::EngineConfig::current_config
```

The documentation for this class was generated from the following files:

- src/engine/engine_config.h
- src/engine/engine_config.cpp

8.9 engine::EngineCreateInfo Struct Reference

```
#include <engine.h>
```

Public Attributes

- std::string game_name
- uint16_t initial_window_width
- uint16_t initial_window_height
- std::string startup_scene

8.9.1 Member Data Documentation

8.9.1.1 game_name

```
std::string engine::EngineCreateInfo::game_name
```

8.9.1.2 initial_window_height

```
uint16_t engine::EngineCreateInfo::initial_window_height
```

8.9.1.3 initial_window_width

```
uint16_t engine::EngineCreateInfo::initial_window_width
```

8.9.1.4 startup_scene

```
std::string engine::EngineCreateInfo::startup_scene
```

The documentation for this struct was generated from the following file:

- src/engine/engine.h

8.10 engine::EngineInstance Class Reference

```
#include <engine.h>
```

Public Member Functions

- EngineInstance (EngineCreateInfo &info)
- ~EngineInstance ()
- void Run ()
- std::string GetVersionString ()
- SceneLoader * GetSceneLoader ()
- OpenGLRenderer * GetRenderer ()
- GLFWwindow * GetWindow ()
- AssetManager * GetAssetManager ()
- PhysicsHandler * GetPhysics ()
- SoundManager * GetSoundManager ()
- void Stop ()
- Component * GetLocalPlayer ()
- void SetLocalPlayer (Component *local_player)
- void Initialize ()
- std::vector< std::string > GetRegisteredComponentsList ()
- Component * CreateComponent (std::string component_type_name)
- Component * LoadComponent (std::string component_type_name, json value)
- void RegisterComponent (std::string type_name, Component *(*factory)(void))
- InputHandler * GetInput ()

8.10.1 Constructor & Destructor Documentation

8.10.1.1 EngineInstance()

```
engine::EngineInstance::EngineInstance (
            EngineCreateInfo & info)
```

8.10.1.2 ~EngineInstance()

```
engine::EngineInstance::~EngineInstance ()
```

8.10.2 Member Function Documentation

8.10.2.1 CreateComponent()

```
Component * engine::EngineInstance::CreateComponent (
            std::string component_type_name)
```

8.10.2.2 GetAssetManager()

```
AssetManager * engine::EngineInstance::GetAssetManager ()
```

8.10.2.3 GetInput()

```
InputHandler * engine::EngineInstance::GetInput ()
```

8.10.2.4 GetLocalPlayer()

```
Component * engine::EngineInstance::GetLocalPlayer ()
```

8.10.2.5 GetPhysics()

```
PhysicsHandler * engine::EngineInstance::GetPhysics ()
```

8.10.2.6 GetRegisteredComponentsList()

```
std::vector< std::string > engine::EngineInstance::GetRegisteredComponentsList ()
```

8.10.2.7 GetRenderer()

```
OpenGLRenderer * engine::EngineInstance::GetRenderer ()
```

8.10.2.8 GetSceneLoader()

```
SceneLoader * engine::EngineInstance::GetSceneLoader ()
```

8.10.2.9 GetSoundManager()

```
SoundManager * engine::EngineInstance::GetSoundManager ()
```

8.10.2.10 GetVersionString()

```
std::string engine::EngineInstance::GetVersionString ()
```

8.10.2.11 GetWindow()

```
GLFWwindow * engine::EngineInstance::GetWindow ()
```

8.10.2.12 Initialize()

```
void engine::EngineInstance::Initialize ()
```

8.10.2.13 LoadComponent()

```
Component * engine::EngineInstance::LoadComponent (
            std::string component_type_name,
            json value)
```

8.10.2.14 RegisterComponent()

```
void engine::EngineInstance::RegisterComponent (
            std::string type_name,
            Component * factory (*)(void))  [inline]
```

8.10.2.15 Run()

```
void engine::EngineInstance::Run ()
```

8.10.2.16 SetLocalPlayer()

```
void engine::EngineInstance::SetLocalPlayer (
            Component * local_player)
```

8.10.2.17 Stop()

```
void engine::EngineInstance::Stop ()
```

The documentation for this class was generated from the following files:

- src/engine/engine.h
- src/engine/engine.cpp

8.11 engine::editor::Gizmo Class Reference

`#include <gizmo.h>`

Public Member Functions

- Gizmo ()
- virtual void draw ()=0

Protected Attributes

- unsigned int vao
- unsigned int vbo

8.11.1 Constructor & Destructor Documentation

8.11.1.1 Gizmo()

```
engine::editor::Gizmo::Gizmo ()
```

8.11.2 Member Function Documentation

8.11.2.1 draw()

```
virtual void engine::editor::Gizmo::draw ()  [pure virtual]
```

8.11.3 Member Data Documentation

8.11.3.1 vao

```
unsigned int engine::editor::Gizmo::vao  [protected]
```

8.11.3.2 vbo

```
unsigned int engine::editor::Gizmo::vbo  [protected]
```

The documentation for this class was generated from the following file:

- src/engine/gizmo.h

8.12 engine::InputHandler Class Reference

`#include <input.h>`

Public Member Functions

- bool KeyDown (int key)
- bool KeyDown (int key, int mods)
- bool KeyUp (int key)
- bool KeyUp (int key, int mods)
- bool KeyPressed (int key)
- bool KeyPressed (int key, int mods)
- bool KeyReleased (int key)
- bool KeyReleased (int key, int mods)
- void DisableCursor ()
- void EnableCursor ()
- bool MouseButtonDown (int button)
- bool MouseButtonPressed (int button)
- bool MouseButtonReleased (int button)
- glm::vec2 MouseDelta ()
- glm::vec2 MousePosition ()
- void Update ()
- void SendInput (int key, int action, int mods)
- void SendMousePosition (glm::vec2 position)
- void SendMouseButton (int button, int action, int mods)

8.12.1 Member Function Documentation

8.12.1.1 DisableCursor()

```
void engine::InputHandler::DisableCursor ()
```

8.12.1.2 EnableCursor()

```
void engine::InputHandler::EnableCursor ()
```

8.12.1.3 KeyDown() [1/2]

```
bool engine::InputHandler::KeyDown (
            int key)
```

8.12.1.4 KeyDown() [2/2]

```
bool engine::InputHandler::KeyDown (
            int key,
            int mods)
```

8.12.1.5 KeyPressed() [1/2]

```
bool engine::InputHandler::KeyPressed (
            int key)
```

8.12.1.6 KeyPressed() [2/2]

```
bool engine::InputHandler::KeyPressed (
            int key,
            int mods)
```

8.12.1.7 KeyReleased() [1/2]

```
bool engine::InputHandler::KeyReleased (
            int key)
```

8.12.1.8 KeyReleased() [2/2]

```
bool engine::InputHandler::KeyReleased (
            int key,
            int mods)
```

8.12.1.9 KeyUp() [1/2]

```
bool engine::InputHandler::KeyUp (
            int key)
```

8.12.1.10 KeyUp() [2/2]

```
bool engine::InputHandler::KeyUp (
            int key,
            int mods)
```

8.12.1.11 MouseButtonDown()

```
bool engine::InputHandler::MouseButtonDown (
            int button)
```

8.12.1.12 MouseButtonPressed()

```
bool engine::InputHandler::MouseButtonPressed (
            int button)
```

8.12.1.13 MouseButtonReleased()

```
bool engine::InputHandler::MouseButtonReleased (
            int button)
```

8.12.1.14 MouseDelta()

```
glm::vec2 engine::InputHandler::MouseDelta ()
```

8.12.1.15 MousePosition()

```
glm::vec2 engine::InputHandler::MousePosition ()
```

8.12.1.16 SendInput()

```
void engine::InputHandler::SendInput (
            int key,
            int action,
            int mods)
```

8.12.1.17 SendMouseButton()

```
void engine::InputHandler::SendMouseButton (
            int button,
            int action,
            int mods)
```

8.12.1.18 SendMousePosition()

```
void engine::InputHandler::SendMousePosition (
            glm::vec2 position)
```

8.12.1.19 Update()

```
void engine::InputHandler::Update ()
```

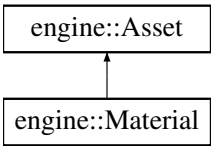
The documentation for this class was generated from the following files:

- src/engine/input.h
- src/engine/input.cpp

8.13 engine::Material Class Reference

```
#include <material.h>
```

Inheritance diagram for engine::Material:



Public Member Functions

- Material (std::string path)
- bool Reload ()
- bool Bind (BaseUniforms base)
- void DrawEditWidget ()

Public Member Functions inherited from engine::Asset

- virtual bool IsValid ()
- std::string GetName ()
- void SetName (std::string name)
- std::string GetPath ()
- void SetPath (std::string path)

Additional Inherited Members

Protected Attributes inherited from engine::Asset

- bool valid = false
- std::string name = "?????"
- std::string path = "INVALID"

8.13.1 Constructor & Destructor Documentation

8.13.1.1 Material()

```
engine::Material::Material (
            std::string path)
```

8.13.2 Member Function Documentation

8.13.2.1 Bind()

```
bool engine::Material::Bind (
            BaseUniforms base)
```

8.13.2.2 DrawEditWidget()

```
void engine::Material::DrawEditWidget ()
```

8.13.2.3 Reload()

```
bool engine::Material::Reload ()
```

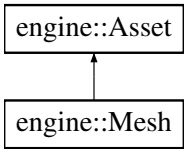
The documentation for this class was generated from the following files:

- src/engine/material.h
- src/engine/material.cpp

8.14 engine::Mesh Class Reference

```
#include <mesh.h>
```

Inheritance diagram for engine::Mesh:



Public Member Functions

- Mesh ()
- ~Mesh ()
- void SetVertices (std::vector< Vertex > v)
- void SetIndices (std::vector< unsigned int > v)
- bool Rebuffer ()
- bool Draw (BaseUniforms base, Material *material)
- bool DrawWireframe (BaseUniforms base, Material *material)

Public Member Functions inherited from engine::Asset

- virtual bool IsValid ()
- std::string GetName ()
- void SetName (std::string name)
- std::string GetPath ()
- void SetPath (std::string path)

Additional Inherited Members

Protected Attributes inherited from engine::Asset

- bool valid = false
- std::string name = "?????"
- std::string path = "INVALID"

8.14.1 Constructor & Destructor Documentation

8.14.1.1 Mesh()

```
engine::Mesh::Mesh ()
```

8.14.1.2 ~Mesh()

```
engine::Mesh::~Mesh ()
```

8.14.2 Member Function Documentation

8.14.2.1 Draw()

```
bool engine::Mesh::Draw (
            BaseUniforms base,
            Material * material)
```

8.14.2.2 DrawWireframe()

```
bool engine::Mesh::DrawWireframe (
            BaseUniforms base,
            Material * material)
```

8.14.2.3 Rebuffer()

```
bool engine::Mesh::Rebuffer ()
```

8.14.2.4 SetIndices()

```
void engine::Mesh::SetIndices (
            std::vector< unsigned int > v)
```

8.14.2.5 SetVertices()

```
void engine::Mesh::SetVertices (
            std::vector< Vertex > v)
```

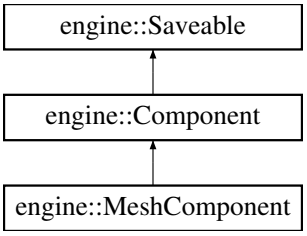
The documentation for this class was generated from the following files:

- src/engine/mesh.h
- src/engine/mesh.cpp

8.15 engine::MeshComponent Class Reference

`#include <mesh_component.h>`

Inheritance diagram for engine::MeshComponent:



Public Member Functions

- virtual void render () override
- virtual void init () override
- virtual void destroy () override
- bool SetMesh (std::string mesh)
- bool SetMaterial (std::string material)
- void DrawWidget () override
- MeshComponent (std::string name)
- MeshComponent (json value)
- bool Load (json value) override
- json Save () override

Public Member Functions inherited from engine::Component

- Component ()
- Component (std::string name)
- virtual ~Component ()
- virtual void begin_play ()
- virtual void end_play ()
- virtual void editor_render ()
- virtual void tick (float dt)
- Transform * GetOwner ()
- void SetOwner (Transform *own)
- std::string GetName ()
- uint64_t GetUID ()
- void ComponentDisplay ()

Additional Inherited Members

Public Attributes inherited from engine::Component

- std::string name
- std::string component_type
- uint64_t uid
- Transform * owner

8.15.1 Constructor & Destructor Documentation

8.15.1.1 MeshComponent() [1/2]

```
engine::MeshComponent::MeshComponent (
            std::string name)
```

8.15.1.2 MeshComponent() [2/2]

```
engine::MeshComponent::MeshComponent (
            json value)
```

8.15.2 Member Function Documentation

8.15.2.1 destroy()

```
void engine::MeshComponent::destroy ()  [override], [virtual]
```

Reimplemented from engine::Component.

8.15.2.2 DrawWidget()

```
void engine::MeshComponent::DrawWidget ()  [override], [virtual]
```

Implements engine::Component.

8.15.2.3 init()

```
void engine::MeshComponent::init ()  [override], [virtual]
```

Reimplemented from engine::Component.

8.15.2.4 Load()

```
bool engine::MeshComponent::Load (
            json value)  [override], [virtual]
```

Implements engine::Component.

8.15.2.5 render()

```
void engine::MeshComponent::render ()  [override], [virtual]
```

Reimplemented from engine::Component.

8.15.2.6 Save()

```
json engine::MeshComponent::Save ()  [override], [virtual]
```

Implements engine::Component.

8.15.2.7 SetMaterial()

```
bool engine::MeshComponent::SetMaterial (
            std::string material)
```

8.15.2.8 SetMesh()

```
bool engine::MeshComponent::SetMesh (
            std::string mesh)
```

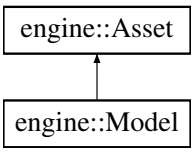
The documentation for this class was generated from the following files:

- src/engine/mesh_component.h
- src/engine/mesh_component.cpp

8.16 engine::Model Class Reference

```
#include <model.h>
```

Inheritance diagram for engine::Model:



Public Member Functions

- Model (std::string path)
- ~Model ()
- std::vector< Mesh * > GetMeshes ()

Public Member Functions inherited from engine::Asset

- virtual bool IsValid ()
- std::string GetName ()
- void SetName (std::string name)
- std::string GetPath ()
- void SetPath (std::string path)

Additional Inherited Members

Protected Attributes inherited from engine::Asset

- bool valid = false
- std::string name = "?????"
- std::string path = "INVALID"

8.16.1 Constructor & Destructor Documentation

8.16.1.1 Model()

```
engine::Model::Model (
            std::string path)
```

8.16.1.2 ~Model()

```
engine::Model::~Model ()
```

8.16.2 Member Function Documentation

8.16.2.1 GetMeshes()

```
std::vector< Mesh * > engine::Model::GetMeshes ()
```

The documentation for this class was generated from the following files:

- src/engine/model.h
- src/engine/model.cpp

8.17 engine::OpenGLRenderer Class Reference

```
#include <opengl_renderer.h>
```

Public Member Functions

- OpenGLRenderer (GLFWwindow **smuggle, struct EngineCreateInfo &create_info)
- ~OpenGLRenderer ()
- bool ShouldClose ()
- float DeltaTime ()
- void StartFrame ()
- void EndFrame ()
- void InitForImGui ()
- int WindowWidth ()
- int WindowHeight ()

8.17.1 Constructor & Destructor Documentation

8.17.1.1 OpenGLRenderer()

```
engine::OpenGLRenderer::OpenGLRenderer (
            GLFWwindow ** smuggle,
            struct EngineCreateInfo & create_info)
```

8.17.1.2 ~OpenGLRenderer()

```
engine::OpenGLRenderer::~OpenGLRenderer ()
```

8.17.2 Member Function Documentation

8.17.2.1 DeltaTime()

```
float engine::OpenGLRenderer::DeltaTime ()
```

8.17.2.2 EndFrame()

```
void engine::OpenGLRenderer::EndFrame ()
```

8.17.2.3 InitForImGui()

```
void engine::OpenGLRenderer::InitForImGui ()
```

8.17.2.4 ShouldClose()

```
bool engine::OpenGLRenderer::ShouldClose ()
```

8.17.2.5 StartFrame()

```
void engine::OpenGLRenderer::StartFrame ()
```

8.17.2.6 WindowHeight()

```
int engine::OpenGLRenderer::WindowHeight ()
```

8.17.2.7 WindowWidth()

```
int engine::OpenGLRenderer::WindowWidth ()
```

The documentation for this class was generated from the following files:

- src/engine/opengl_renderer.h
- src/engine/opengl_renderer.cpp

8.18 engine::PhysicsHandler Class Reference

```
#include <physics.h>
```

Public Member Functions

- bool CheckTrigger (CollisionComponent ∗collider, int trigger_group)
- void RegisterCollider (CollisionComponent ∗collider)
- void DeregisterCollider (CollisionComponent ∗collider)
- std::vector< CollisionComponent ∗ > ∗ GetColliders ()
- void RegisterTrigger (TriggerComponent ∗trigger)
- void DeregisterTrigger (TriggerComponent ∗trigger)
- std::vector< TriggerComponent ∗ > ∗ GetTriggers ()

8.18.1 Member Function Documentation

8.18.1.1 CheckTrigger()

```
bool engine::PhysicsHandler::CheckTrigger (
            CollisionComponent * collider,
            int trigger_group)
```

8.18.1.2 DeregisterCollider()

```
void engine::PhysicsHandler::DeregisterCollider (
            CollisionComponent * collider)
```

8.18.1.3 DeregisterTrigger()

```
void engine::PhysicsHandler::DeregisterTrigger (
            TriggerComponent * trigger)
```

8.18.1.4 GetColliders()

```
std::vector< CollisionComponent * > * engine::PhysicsHandler::GetColliders ()
```

8.18.1.5 GetTriggers()

```
std::vector< TriggerComponent * > * engine::PhysicsHandler::GetTriggers ()
```

8.18.1.6 RegisterCollider()

```
void engine::PhysicsHandler::RegisterCollider (
            CollisionComponent * collider)
```

8.18.1.7 RegisterTrigger()

```
void engine::PhysicsHandler::RegisterTrigger (
            TriggerComponent * trigger)
```

The documentation for this class was generated from the following files:

- src/engine/physics.h
- src/engine/physics.cpp

8.19 engine::Ray Struct Reference

```
#include <physics.h>
```

Public Attributes

- glm::vec3 position
- glm::vec3 direction

8.19.1 Member Data Documentation

8.19.1.1 direction

```
glm::vec3 engine::Ray::direction
```

8.19.1.2 position

```
glm::vec3 engine::Ray::position
```

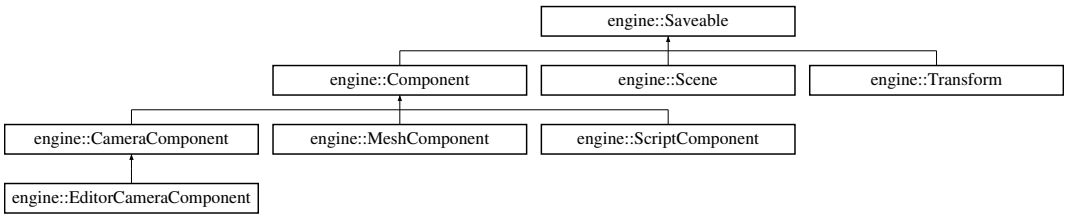
The documentation for this struct was generated from the following file:

- src/engine/physics.h

8.20 engine::Saveable Class Reference

`#include <saveable.h>`

Inheritance diagram for engine::Saveable:



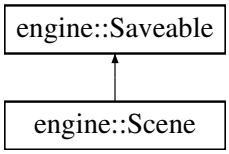
The documentation for this class was generated from the following file:

- src/engine/saveable.h

8.21 engine::Scene Class Reference

`#include <scene.h>`

Inheritance diagram for engine::Scene:



Public Member Functions

- Scene ()
- Scene (std::string name)
- ~Scene ()
- void Render ()
- void Tick (float dt)
- void EditorRender ()
- json Save () override
- bool Load (json value) override
- Transform * GetRoot ()
- bool IsValid ()
- void SetCamera (CameraComponent *cam)
- void SetMainCamera (CameraComponent *cam)
- void SetCameraToMain ()
- Scene * Clone ()
- glm::mat4 GetCurrentCameraMatrix ()
- glm::vec3 GetCurrentCameraPosition ()
- CameraComponent * GetCurrentCamera ()

8.21.1 Constructor & Destructor Documentation

8.21.1.1 Scene() [1/2]

```
engine::Scene::Scene ()
```

8.21.1.2 Scene() [2/2]

```
engine::Scene::Scene (
            std::string name)
```

8.21.1.3 ~Scene()

```
engine::Scene::~Scene ()
```

8.21.2 Member Function Documentation

8.21.2.1 Clone()

```
Scene * engine::Scene::Clone ()
```

8.21.2.2 EditorRender()

```
void engine::Scene::EditorRender ()
```

8.21.2.3 GetCurrentCamera()

```
CameraComponent * engine::Scene::GetCurrentCamera ()
```

8.21.2.4 GetCurrentCameraMatrix()

```
glm::mat4 engine::Scene::GetCurrentCameraMatrix ()
```

8.21.2.5 GetCurrentCameraPosition()

```
glm::vec3 engine::Scene::GetCurrentCameraPosition ()
```

8.21.2.6 GetRoot()

```
Transform * engine::Scene::GetRoot ()
```

8.21.2.7 IsValid()

```
bool engine::Scene::IsValid ()
```

8.21.2.8 Load()

```
bool engine::Scene::Load (
            json value)  [override], [virtual]
```

Implements engine::Saveable.

8.21.2.9 Render()

```
void engine::Scene::Render ()
```

8.21.2.10 Save()

```
json engine::Scene::Save ()  [override], [virtual]
```

Implements engine::Saveable.

8.21.2.11 SetCamera()

```
void engine::Scene::SetCamera (
            CameraComponent * cam)
```

8.21.2.12 SetCameraToMain()

```
void engine::Scene::SetCameraToMain ()
```

8.21.2.13 SetMainCamera()

```
void engine::Scene::SetMainCamera (
            CameraComponent * cam)
```

8.21.2.14 Tick()

```
void engine::Scene::Tick (
            float dt)
```

The documentation for this class was generated from the following files:

- src/engine/scene.h
- src/engine/scene.cpp

8.22 engine::SceneLoader Class Reference

`#include <scene_loader.h>`

Public Member Functions

- SceneLoader ()
- ~SceneLoader ()
- bool LoadScene (std::string path, void *data)
- void LoadDefaultScene ()
- void SaveScene (std::string path)
- void DeloadScene ()
- bool UpdateCurrentScene (float dt)
- bool RenderCurrentScene ()
- Scene * GetCurrentScene ()
- void SaveAsCheckpoint ()
- void RewindToCheckpoint ()
- void * GetPersistentData ()

8.22.1 Constructor & Destructor Documentation

8.22.1.1 SceneLoader()

```
engine::SceneLoader::SceneLoader ()
```

8.22.1.2 ~SceneLoader()

```
engine::SceneLoader::~SceneLoader ()
```

8.22.2 Member Function Documentation

8.22.2.1 DeloadScene()

```
void engine::SceneLoader::DeloadScene ()
```

8.22.2.2 GetCurrentScene()

```
Scene * engine::SceneLoader::GetCurrentScene ()
```

8.22.2.3 GetPersistentData()

```
void * engine::SceneLoader::GetPersistentData ()
```

8.22.2.4 LoadDefaultScene()

```
void engine::SceneLoader::LoadDefaultScene ()
```

8.22.2.5 LoadScene()

```
bool engine::SceneLoader::LoadScene (
            std::string path,
            void * data)
```

8.22.2.6 RenderCurrentScene()

```
bool engine::SceneLoader::RenderCurrentScene ()
```

8.22.2.7 RewindToCheckpoint()

```
void engine::SceneLoader::RewindToCheckpoint ()
```

8.22.2.8 SaveAsCheckpoint()

```
void engine::SceneLoader::SaveAsCheckpoint ()
```

8.22.2.9 SaveScene()

```
void engine::SceneLoader::SaveScene (
            std::string path)
```

8.22.2.10 UpdateCurrentScene()

```
bool engine::SceneLoader::UpdateCurrentScene (
            float dt)
```

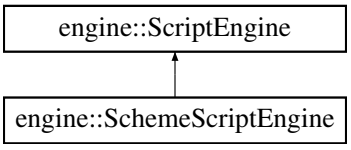
The documentation for this class was generated from the following files:

- src/engine/scene_loader.h
- src/engine/scene_loader.cpp

8.23 engine::SchemeScriptEngine Class Reference

```
#include <script_engine.h>
```

Inheritance diagram for engine::SchemeScriptEngine:



Public Member Functions

- SchemeScriptEngine ()
- ~SchemeScriptEngine ()
- void eval_string () override

8.23.1 Constructor & Destructor Documentation

8.23.1.1 SchemeScriptEngine()

```
engine::SchemeScriptEngine::SchemeScriptEngine ()
```

8.23.1.2 ~SchemeScriptEngine()

```
engine::SchemeScriptEngine::~SchemeScriptEngine ()
```

8.23.2 Member Function Documentation

8.23.2.1 eval_string()

```
void engine::SchemeScriptEngine::eval_string ()  [override], [virtual]
```

Implements engine::ScriptEngine.

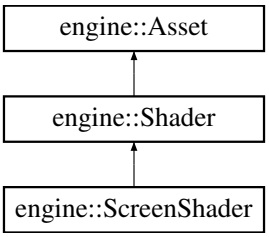
The documentation for this class was generated from the following file:

- src/engine/script_engine.h

8.24 engine::ScreenShader Class Reference

```
#include <screen_shader.h>
```

Inheritance diagram for engine::ScreenShader:



Additional Inherited Members

Public Member Functions inherited from engine::Shader

- Shader (std::string path)
- ~Shader ()
- bool SetUniforms (UniformList &u)
- bool SetUniform (std::string key, Uniform u)
- void Reload ()
- bool Bind ()

Public Member Functions inherited from engine::Asset

- virtual bool IsValid ()
- std::string GetName ()
- void SetName (std::string name)
- std::string GetPath ()
- void SetPath (std::string path)

Protected Attributes inherited from engine::Shader

- unsigned int shader_handle = 0
- std::string vertex_src_str
- std::string fragment_src_str

Protected Attributes inherited from engine::Asset

- bool valid = false
- std::string name = "?????"
- std::string path = "INVALID"

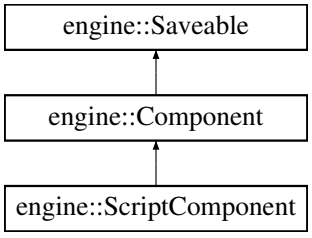
The documentation for this class was generated from the following files:

- src/engine/screen_shader.h
- src/engine/screen_shader.cpp

8.25 engine::ScriptComponent Class Reference

`#include <script_component.h>`

Inheritance diagram for engine::ScriptComponent:



Public Member Functions

- virtual void tick (float dt) override
- virtual void init () override
- virtual void destroy () override
- void ReloadScripts ()

Public Member Functions inherited from engine::Component

- Component ()
- Component (std::string name)
- virtual ~Component ()
- virtual void begin_play ()
- virtual void end_play ()
- virtual void render ()
- virtual void editor_render ()
- virtual void DrawWidget ()=0
- Transform * GetOwner ()
- void SetOwner (Transform *own)
- std::string GetName ()
- virtual json Save () override=0
- virtual bool Load (json value) override=0
- uint64_t GetUID ()
- void ComponentDisplay ()

Additional Inherited Members

Public Attributes inherited from engine::Component

- std::string name
- std::string component_type
- uint64_t uid
- Transform * owner

8.25.1 Member Function Documentation

8.25.1.1 destroy()

```
virtual void engine::ScriptComponent::destroy ()  [override], [virtual]
```

Reimplemented from engine::Component.

8.25.1.2 init()

```
virtual void engine::ScriptComponent::init ()  [override], [virtual]
```

Reimplemented from engine::Component.

8.25.1.3 ReloadScripts()

```
void engine::ScriptComponent::ReloadScripts ()
```

8.25.1.4 tick()

```
virtual void engine::ScriptComponent::tick (
            float dt)  [override], [virtual]
```

Reimplemented from engine::Component.

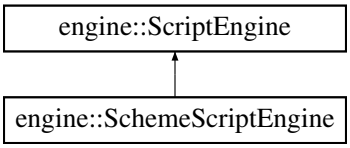
The documentation for this class was generated from the following files:

- src/engine/script_component.h
- src/engine/script_component.cpp

8.26 engine::ScriptEngine Class Reference

```
#include <script_engine.h>
```

Inheritance diagram for engine::ScriptEngine:



Public Member Functions

- virtual void eval_string ()=0

8.26.1 Member Function Documentation

8.26.1.1 eval_string()

```
virtual void engine::ScriptEngine::eval_string ()  [pure virtual]
```

Implemented in engine::SchemeScriptEngine.

The documentation for this class was generated from the following file:

- src/engine/script_engine.h

8.27 engine::ScriptFile Struct Reference

`#include <script_engine.h>`

Public Attributes

- std::string script_path
- std::string cached_script
- bool stale

8.27.1 Member Data Documentation

8.27.1.1 cached_script

`std::string engine::ScriptFile::cached_script`

8.27.1.2 script_path

`std::string engine::ScriptFile::script_path`

8.27.1.3 stale

`bool engine::ScriptFile::stale`

The documentation for this struct was generated from the following file:

- src/engine/script_engine.h

8.28 engine::ScriptLiteral Struct Reference

`#include <script_engine.h>`

Public Attributes

- std::string script

8.28.1 Member Data Documentation

8.28.1.1 script

`std::string engine::ScriptLiteral::script`

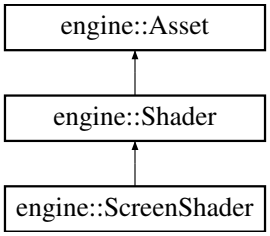
The documentation for this struct was generated from the following file:

- src/engine/script_engine.h

8.29 engine::Shader Class Reference

`#include <shader.h>`

Inheritance diagram for engine::Shader:



Public Member Functions

- Shader (std::string path)
- ~Shader ()
- bool SetUniforms (UniformList &u)
- bool SetUniform (std::string key, Uniform u)
- void Reload ()
- bool Bind ()

Public Member Functions inherited from engine::Asset

- virtual bool IsValid ()
- std::string GetName ()
- void SetName (std::string name)
- std::string GetPath ()
- void SetPath (std::string path)

Protected Attributes

- unsigned int shader_handle = 0
- std::string vertex_src_str
- std::string fragment_src_str

Protected Attributes inherited from engine::Asset

- bool valid = false
- std::string name = "?????"
- std::string path = "INVALID"

8.29.1 Constructor & Destructor Documentation

8.29.1.1 Shader()

```
engine::Shader::Shader (
            std::string path)
```

8.29.1.2 ~Shader()

```
engine::Shader::~Shader ()
```

8.29.2 Member Function Documentation

8.29.2.1 Bind()

```
bool engine::Shader::Bind ()
```

8.29.2.2 Reload()

```
void engine::Shader::Reload ()
```

8.29.2.3 SetUniform()

```
bool engine::Shader::SetUniform (
            std::string key,
            Uniform u)
```

8.29.2.4 SetUniforms()

```
bool engine::Shader::SetUniforms (
            UniformList & u)
```

8.29.3 Member Data Documentation

8.29.3.1 fragment_src_str

```
std::string engine::Shader::fragment_src_str  [protected]
```

8.29.3.2 shader_handle

```
unsigned int engine::Shader::shader_handle = 0  [protected]
```

8.29.3.3 vertex_src_str

```
std::string engine::Shader::vertex_src_str  [protected]
```

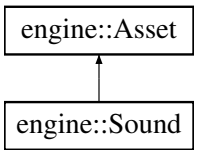
The documentation for this class was generated from the following files:

- src/engine/shader.h
- src/engine/shader.cpp

8.30 engine::Sound Class Reference

`#include <sound.h>`

Inheritance diagram for engine::Sound:



Public Member Functions

- Sound (std::string path)
- void PlayLooped ()

Public Member Functions inherited from engine::Asset

- virtual bool IsValid ()
- std::string GetName ()
- void SetName (std::string name)
- std::string GetPath ()
- void SetPath (std::string path)

Additional Inherited Members

Protected Attributes inherited from engine::Asset

- bool valid = false
- std::string name = "?????"
- std::string path = "INVALID"

8.30.1 Constructor & Destructor Documentation

8.30.1.1 Sound()

```
engine::Sound::Sound (
            std::string path)
```

8.30.2 Member Function Documentation

8.30.2.1 PlayLooped()

```
void engine::Sound::PlayLooped ()
```

The documentation for this class was generated from the following files:

- src/engine/sound.h
- src/engine/sound.cpp

8.31 engine::SoundManager Class Reference

`#include <sound_manager.h>`

Public Member Functions

- SoundManager ()
- ~SoundManager ()
- void PlaySound (std::string sound)
- ma_sound * LoadSound (std::string path)

8.31.1 Constructor & Destructor Documentation

8.31.1.1 SoundManager()

```
engine::SoundManager::SoundManager ()
```

8.31.1.2 ~SoundManager()

```
engine::SoundManager::~SoundManager ()
```

8.31.2 Member Function Documentation

8.31.2.1 LoadSound()

```
ma_sound * engine::SoundManager::LoadSound (
            std::string path)
```

8.31.2.2 PlaySound()

```
void engine::SoundManager::PlaySound (
            std::string sound)
```

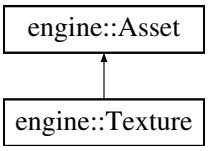
The documentation for this class was generated from the following files:

- src/engine/sound_manager.h
- src/engine/sound_manager.cpp

8.32 engine::Texture Class Reference

`#include <texture.h>`

Inheritance diagram for engine::Texture:



Public Member Functions

- Texture (std::string path)
- ~Texture ()
- void Bind ()
- unsigned int GetTextureHandle ()

Public Member Functions inherited from engine::Asset

- virtual bool IsValid ()
- std::string GetName ()
- void SetName (std::string name)
- std::string GetPath ()
- void SetPath (std::string path)

Additional Inherited Members

Protected Attributes inherited from engine::Asset

- bool valid = false
- std::string name = "?????"
- std::string path = "INVALID"

8.32.1 Constructor & Destructor Documentation

8.32.1.1 Texture()

```
engine::Texture::Texture (
            std::string path)
```

8.32.1.2 ~Texture()

```
engine::Texture::~Texture ()
```

8.32.2 Member Function Documentation

8.32.2.1 Bind()

```
void engine::Texture::Bind ()
```

8.32.2.2 GetTextureHandle()

```
unsigned int engine::Texture::GetTextureHandle ()
```

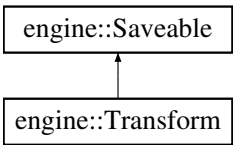
The documentation for this class was generated from the following files:

- src/engine/texture.h
- src/engine/texture.cpp

8.33 engine::Transform Class Reference

```
#include <transform.h>
```

Inheritance diagram for engine::Transform:



Public Member Functions

- Transform ()
- Transform (json value)
- Transform (std::string name)
- ~Transform ()
- void SetName (std::string name)
- json Save () override
- bool Load (json value) override
- void ProcessEditorRender ()
- void ProcessRender ()
- void ProcessTick (float dt)
- void ProcessInit ()
- void AddComponent (Component *component)
- Component * GetComponent (std::string name)
- void AddChild (Transform *tr)
- Transform * FindTransformByUID (uint64_t id)
- Component * FindComponentByUID (uint64_t id)
- Transform * FindChildByName (std::string name)
- Transform * RemoveChild (Transform *tr)
- std::vector< Transform * > GetChildren ()
- void Translate (glm::vec3 offset)

- void Scale (glm::vec3 amount)
- void Rotate (glm::vec3 axis, float amount)
- glm::vec3 Forward ()
- glm::vec3 Backward ()
- glm::vec3 Up ()
- glm::vec3 Down ()
- glm::vec3 Left ()
- glm::vec3 Right ()
- glm::mat4 GetModelMatrix ()
- glm::mat4 GetModelMatrixUnscaled ()
- glm::vec3 GetGlobalPosition ()
- glm::vec3 GetLocalPosition ()
- glm::quat GetGlobalRotation ()
- glm::quat GetLocalRotation ()
- glm::vec3 GetGlobalScale ()
- glm::vec3 GetLocalScale ()
- std::string GetName ()
- uint64_t GetUID ()
- void DrawWidget ()

Public Attributes

- glm::vec3 position
- glm::quat rotation
- glm::vec3 scale
- std::string name
- uint64_t uid
- glm::vec3 rotationEuler
- Transform * parent = nullptr
- std::vector< Transform * > children
- std::vector< Component * > components

8.33.1 Constructor & Destructor Documentation

8.33.1.1 Transform() [1/3]

```
engine::Transform::Transform ()
```

Default constructor

8.33.1.2 Transform() [2/3]

```
engine::Transform::Transform (
            json value)
```

"load" constructor

Called by the level loader to deserialize transforms from json

Parameters

value	The JSON to deserialize

8.33.1.3 Transform() [3/3]

```
engine::Transform::Transform (
            std::string name)
```

Same as default constructor, but with the node name set

Parameters

name	The desired node name

8.33.1.4 ~Transform()

```
engine::Transform::~Transform ()
```

Destroys this transform, all of its components, and then recursively destroys its children

8.33.2 Member Function Documentation

8.33.2.1 AddChild()

```
void engine::Transform::AddChild (
            Transform * tr)
```

Add a child transform to this transform

If the transform is already a child of this transform log a warning and ignore it

Parameters

tr	the transform to add as a child

8.33.2.2 AddComponent()

```
void engine::Transform::AddComponent (
            Component * component)
```

Add a component to this transform

Parameters

component	A pointer to the component to add

8.33.2.3 Backward()

```
glm::vec3 engine::Transform::Backward ()
```

Returns the backward direction of this transform

Returns

the backward direction of the transform

8.33.2.4 Down()

```
glm::vec3 engine::Transform::Down ()
```

Returns the down direction of this transform

Returns

the down direction of the transform

8.33.2.5 DrawWidget()

```
void engine::Transform::DrawWidget ()
```

Draws the editor UI for this transform

8.33.2.6 FindChildByName()

```
Transform * engine::Transform::FindChildByName (
            std::string name)
```

Return the child transform with the given name

If more than one child has the given name, return the first in the array

If no child has the given name, return nullptr

Parameters

name	The name of the child to return

Returns

A pointer to the transform with the given name, or nullptr

8.33.2.7 FindComponentByUID()

```
Component * engine::Transform::FindComponentByUID (
            uint64_t id)
```

Return the attached component that has the given id

If no component has the given ID, return nullptr

Parameters

id	The id of the component to return

Returns

A pointer to the component with the given id, or nullptr

8.33.2.8 FindTransformByUID()

```
Transform * engine::Transform::FindTransformByUID (
            uint64_t id)
```

Return the child transform that has the given id

If no child has the given ID, return nullptr

Parameters

id	The id of the transform to return

Returns

A pointer to the transform with the given id, or nullptr

8.33.2.9 Forward()

```
glm::vec3 engine::Transform::Forward ()
```

Returns the forward direction of this transform

Returns

the forward direction of the transform

8.33.2.10 GetChildren()

```
std::vector< Transform * > engine::Transform::GetChildren ()
```

Return the children array of this transform

Returns

The children array of this transform

8.33.2.11 GetComponent()

```
Component * engine::Transform::GetComponent (
            std::string name)
```

Get a component on this transform by name

If multiple components have the same name, the first in the array is returned

If no component has the given name, return nullptr

Parameters

name	The name of the component to return

Returns

A pointer to the component with the given name, or nullptr

8.33.2.12 GetGlobalPosition()

```
glm::vec3 engine::Transform::GetGlobalPosition ()
```

Calculates the position of this transform in global coordinates

Returns

the global position vector of this transform

8.33.2.13 GetGlobalRotation()

```
glm::quat engine::Transform::GetGlobalRotation ()
```

Returns the rotation of this transform in global coordinates

Returns

the quaternion rotation of this transform in global coordinates

8.33.2.14 GetGlobalScale()

```
glm::vec3 engine::Transform::GetGlobalScale ()
```

Returns the scale of this transform in global coordinates

Returns

the scale value of this transform in global coordinates

8.33.2.15 GetLocalPosition()

```
glm::vec3 engine::Transform::GetLocalPosition ()
```

Returns the position of this transform in local coordinates

Returns

the local position vector of this transform

8.33.2.16 GetLocalRotation()

```
glm::quat engine::Transform::GetLocalRotation ()
```

Returns the rotation of this transform in local coordinates

Returns

the quaternion rotation of this transform in local coordinates

8.33.2.17 GetLocalScale()

```
glm::vec3 engine::Transform::GetLocalScale ()
```

Returns the scale of this transform in local coordinates

Returns

the scale value of this transform in local coordinates

8.33.2.18 GetModelMatrix()

```
glm::mat4 engine::Transform::GetModelMatrix ()
```

Calculate the model matrix given the instances position, rotation, and scale

Returns

the model matrix corresponding to this instance

8.33.2.19 GetModelMatrixUnscaled()

```
glm::mat4 engine::Transform::GetModelMatrixUnscaled ()
```

Calculates the model matrix with the scale normalized to 1

This is used internally for some editor functionality.

Returns

The model matrix corresponding to this instance with a unit scale

8.33.2.20 GetName()

```
std::string engine::Transform::GetName ()
```

Returns the name of this transform

Returns

the name of this transform

8.33.2.21 GetUID()

```
uint64_t engine::Transform::GetUID ()
```

Returns the uid of this transform

Returns

the uid of this transform

8.33.2.22 Left()

```
glm::vec3 engine::Transform::Left ()
```

Returns the left direction of this transform

Returns

the left direction of the transform

8.33.2.23 Load()

```
bool engine::Transform::Load (
            json value)  [override], [virtual]
```

Load the transform properties saved with `Save()` into the transform

Returns

true on success, false on failure

Implements engine::Saveable.

8.33.2.24 ProcessEditorRender()

```
void engine::Transform::ProcessEditorRender ()
```

This is the render function called by the editor

8.33.2.25 ProcessInit()

```
void engine::Transform::ProcessInit ()
```

This is the initialization function called to initialize components.

8.33.2.26 ProcessRender()

```
void engine::Transform::ProcessRender ()
```

This is the render function called by the game

8.33.2.27 ProcessTick()

```
void engine::Transform::ProcessTick (
            float dt)
```

Runs once per frame when the game simulation is running

8.33.2.28 RemoveChild()

```
Transform * engine::Transform::RemoveChild (
            Transform * tr)
```

Removes the given transform from the children array, and returns it

Does not automatically destroy the transform

If the given transform does not exist in the child array, return nullptr

Parameters

tr	The transform to remove

Returns

The transform that was removed

8.33.2.29 Right()

```
glm::vec3 engine::Transform::Right ()
```

Returns the right direction of this transform

Returns

the right direction of the transform

8.33.2.30 Rotate()

```
void engine::Transform::Rotate (
            glm::vec3 axis,
            float amount)
```

Rotates the transform along the given axis by the given amount

Parameters

axis	The axis to rotate around
amount	The rotation amount, in degrees

8.33.2.31 Save()

```
json engine::Transform::Save ()  [override], [virtual]
```

Save's the transform's properties to a JSON object

Returns

the JSON object representing the transform

Implements engine::Saveable.

8.33.2.32 Scale()

```
void engine::Transform::Scale (
            glm::vec3 amount)
```

Scales this transform by the given multiplier

Parameters

amount	The vector to multiply the scale by

8.33.2.33 SetName()

```
void engine::Transform::SetName (
            std::string name)
```

Sets the node's name

Parameters

name	The desired node name

8.33.2.34 Translate()

```
void engine::Transform::Translate (
            glm::vec3 offset)
```

Translates this transform in local coordinates

Parameters

offset	The vector to translate the transform by

8.33.2.35 Up()

```
glm::vec3 engine::Transform::Up ()
```

Returns the up direction of this transform

Returns

the up direction of the transform

8.33.3 Member Data Documentation

8.33.3.1 children

```
std::vector<Transform *> engine::Transform::children
```

8.33.3.2 components

```
std::vector<Component*> engine::Transform::components
```

8.33.3.3 name

```
std::string engine::Transform::name
```

8.33.3.4 parent

```
Transform* engine::Transform::parent = nullptr
```

8.33.3.5 position

```
glm::vec3 engine::Transform::position
```

8.33.3.6 rotation

```
glm::quat engine::Transform::rotation
```

8.33.3.7 rotationEuler

```
glm::vec3 engine::Transform::rotationEuler
```

8.33.3.8 scale

```
glm::vec3 engine::Transform::scale
```

8.33.3.9 uid

```
uint64_t engine::Transform::uid
```

The documentation for this class was generated from the following files:

- src/engine/transform.h
- src/engine/transform.cpp

8.34 engine::Uniform_d Struct Reference

```
#include <shader.h>
```

Public Attributes

- double value

8.34.1 Member Data Documentation

8.34.1.1 value

```
double engine::Uniform_d::value
```

The documentation for this struct was generated from the following file:

- src/engine/shader.h

8.35 engine::Uniform_f Struct Reference

```
#include <shader.h>
```

Public Member Functions

- Uniform_f (float f)

Public Attributes

- float value

8.35.1 Constructor & Destructor Documentation

8.35.1.1 Uniform_f()

```
engine::Uniform_f::Uniform_f (
            float f)  [inline]
```

8.35.2 Member Data Documentation

8.35.2.1 value

```
float engine::Uniform_f::value
```

The documentation for this struct was generated from the following file:

- src/engine/shader.h

8.36 engine::Uniform_i Struct Reference

`#include <shader.h>`

Public Member Functions

- Uniform_i (int v)

Public Attributes

- int value

8.36.1 Constructor & Destructor Documentation

8.36.1.1 Uniform_i()

```
engine::Uniform_i::Uniform_i (
            int v)  [inline]
```

8.36.2 Member Data Documentation

8.36.2.1 value

```
int engine::Uniform_i::value
```

The documentation for this struct was generated from the following file:

- src/engine/shader.h

8.37 engine::Uniform_mat4 Struct Reference

`#include <shader.h>`

Public Member Functions

- Uniform_mat4 (glm::mat4 m)

Public Attributes

- glm::mat4 value

8.37.1 Constructor & Destructor Documentation

8.37.1.1 Uniform_mat4()

```
engine::Uniform_mat4::Uniform_mat4 (
            glm::mat4 m)  [inline]
```

8.37.2 Member Data Documentation

8.37.2.1 value

```
glm::mat4 engine::Uniform_mat4::value
```

The documentation for this struct was generated from the following file:

- src/engine/shader.h

8.38 engine::Uniform_ui Struct Reference

```
#include <shader.h>
```

Public Attributes

- unsigned int value

8.38.1 Member Data Documentation

8.38.1.1 value

```
unsigned int engine::Uniform_ui::value
```

The documentation for this struct was generated from the following file:

- src/engine/shader.h

8.39 engine::Uniform_vec3 Struct Reference

```
#include <shader.h>
```

Public Member Functions

- Uniform_vec3 (glm::vec3 v)

Public Attributes

- glm::vec3 value

8.39.1 Constructor & Destructor Documentation

8.39.1.1 Uniform_vec3()

```
engine::Uniform_vec3::Uniform_vec3 (
            glm::vec3 v)  [inline]
```

8.39.2 Member Data Documentation

8.39.2.1 value

```
glm::vec3 engine::Uniform_vec3::value
```

The documentation for this struct was generated from the following file:

- src/engine/shader.h

8.40 engine::UniformList Class Reference

```
#include <shader.h>
```

Public Attributes

- std::map< std::string, Uniform > uniforms

8.40.1 Member Data Documentation

8.40.1.1 uniforms

```
std::map<std::string,Uniform> engine::UniformList::uniforms
```

The documentation for this class was generated from the following file:

- src/engine/shader.h

8.41 Vertex Struct Reference

```
#include <mesh.h>
```

Public Attributes

- glm::vec3 position
- glm::vec3 normal
- glm::vec2 t_coord

8.41.1 Member Data Documentation

8.41.1.1 normal

```
glm::vec3 Vertex::normal
```

8.41.1.2 position

```
glm::vec3 Vertex::position
```

8.41.1.3 t_coord

```
glm::vec2 Vertex::t_coord
```

The documentation for this struct was generated from the following file:

- src/engine/mesh.h

Chapter 9

File Documentation

9.1 src/engine/asset.cpp File Reference

```
#include "asset.h"
```

Namespaces

- namespace engine

9.2 src/engine/asset.h File Reference

```
#include <string>
```

Classes

- class engine::Asset

Namespaces

- namespace engine

9.3 asset.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include <string>
00003
00004 namespace engine {
00005
00006
00007 class Asset {
00008 public:
00009   virtual bool IsValid();
00010   std::string GetName();
00011   void SetName(std::string name);
00012   std::string GetPath();
00013   void SetPath(std::string path);
00014
00015 protected:
00016   bool valid = false;
00017   std::string name = "?????";
00018   std::string path = "INVALID";
00019 };
00020
00021
00022
00023 }
```

9.4 src/engine/asset_manager.cpp File Reference

```
#include "asset_manager.h"
#include "engine.h"
#include <filesystem>
```

Namespaces

- namespace engine

Functions

- std::string engine::trim_str (std::string in)

9.5 src/engine/asset_manager.h File Reference

```
#include "../json/json.hpp"
#include <map>
#include "model.h"
#include "material.h"
#include "texture.h"
#include "shader.h"
#include "scene.h"
#include "sound.h"
```

Classes

- class engine::AssetManager

Namespaces

- namespace engine

Typedefs

- using json = nlohmann::json

9.5.1 Typedef Documentation

9.5.1.1 json

```
using json = nlohmann::json
```

9.6 asset_manager.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "../json/json.hpp"
00003 #include <map>
00004
00005 #include "model.h"
00006 #include "material.h"
00007 #include "texture.h"
00008 #include "shader.h"
00009 #include "scene.h"
00010 #include "sound.h"
00011
00012 using json = nlohmann::json;
00013
00014 namespace engine {
00015
00016
00017 class AssetManager {
00018 public:
00019   void Rescan();
00020   Mesh *GetMeshOrNull(std::string key);
00021
00022   Material* GetMaterialOrNull(std::string key);
00023   Sound* GetSoundOrNull(std::string key);
00024   Shader* GetShaderOrNull(std::string key);
00025   Texture* GetTextureOrNull(std::string key);
00026
00027   std::map<std::string, Model*> loaded_models;            //.fbx
00028   std::map<std::string, Mesh*> loaded_meshes;
00029   std::map<std::string, Material*> loaded_materials;      //.mtl
00030   std::map<std::string, Texture*> loaded_textures;        //.png
00031   std::map<std::string, Shader *> loaded_shaders;         //.shd
00032   std::map<std::string, Sound*> loaded_sounds;
00033 private:
00034   void ProcessModel(std::filesystem::path pth);
00035   void ProcessMaterial(std::filesystem::path pth);
00036   void ProcessShader(std::filesystem::path pth);
00037   void ProcessTexture(std::filesystem::path pth);
00038   void ProcessSound(std::filesystem::path pth);
00039 };
00040
00041 }
```

9.7 src/engine/camera_component.cpp File Reference

```
#include "camera_component.h"
#include <glm/ext/matrix_transform.hpp>
#include "engine.h"
#include "input.h"
#include <glm/geometric.hpp>
#include <glm/gtc/matrix_inverse.hpp>
#include <string>
```

Namespaces

- namespace engine

Functions

- engine::FACTORY_DEF (EditorCameraComponent)
- engine::FACTORY_DEF (CameraComponent)

9.8 src/engine/camera_component.h File Reference

```
#include "../json/json.hpp"
#include "component.h"
#include <string>
#include "physics.h"
```

Classes

- class engine::CameraComponent
- class engine::EditorCameraComponent

Namespaces

- namespace engine

Functions

- engine::FACTORY (EditorCameraComponent)
- engine::FACTORY (CameraComponent)

9.9 camera_component.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "../json/json.hpp"
00003 #include "component.h"
00004 #include <string>
00005 #include "physics.h"
00006
00007 namespace engine {
00008
00009 FACTORY(EditorCameraComponent);
00010 FACTORY(CameraComponent);
00011
00012
00013 struct Ray;
00014
00015
00016
00017 class CameraComponent : public Component {
00018 public:
00019   CameraComponent();
00020   CameraComponent(std::string name);
00021   virtual ~CameraComponent();
00022   virtual void tick(float dt) override;
00023   virtual glm::mat4 GetCameraMatrix();
00024   virtual Ray ScreenPointToRay(glm::vec2 screen_point);
00025   virtual json Save() override;
00026   virtual bool Load(json value) override;
00027   virtual void DrawWidget() override;
00028 };
00029
00030 class EditorCameraComponent : public CameraComponent {
00031 public:
00032   EditorCameraComponent();
00033   EditorCameraComponent(std::string name);
00034   virtual ~EditorCameraComponent();
00035   virtual void tick(float dt) override;
00036   virtual void DrawWidget() override;
00037
00038   virtual json Save() override;
00039   virtual bool Load(json value) override;
00040
00041
00042
00043
00044 protected:
00045   float movement_speed = 25.0f;
00046 };
00047
00048
00049 }
```

9.10 src/engine/component.cpp File Reference

```
#include "component.h"
#include "util.h"
#include "../imgui/imgui.h"
```

Namespaces

- namespace engine

9.11 src/engine/component.h File Reference

```
#include "transform.h"
#include "saveable.h"
#include <string>
#include "../imgui/imgui.h"
```

Classes

- class engine::Component

Namespaces

- namespace engine

Macros

- #define FACTORY(type)
- #define FACTORY_DEF(type)
- #define FACTORY_REF(type)

9.11.1 Macro Definition Documentation

9.11.1.1 FACTORY

```
#define FACTORY(
                type)
```

Value:

```
engine::Component *__Get ## type()
```

9.11.1.2 FACTORY_DEF

```
#define FACTORY_DEF(
                type)
```

Value:

```
engine ::Component *__Get##type() {                                    \
  return (engine::Component *)(new type());                           \
}                                                                      \
```

9.11.1.3 FACTORY_REF

```
#define FACTORY_REF(
                type)
```

Value:

```
__Get ## type
```

9.12 component.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "transform.h"
00003 #include "saveable.h"
00004 #include <string>
00005 #include "../imgui/imgui.h"
00006
00007
00008 #define FACTORY(type) engine::Component *__Get ## type()
00009
00010 #define FACTORY_DEF(type)                                                    \
00011   engine ::Component *__Get##type() {                                         \
00012     return (engine::Component *)(new type());                                \
00013   }                                                                           \
00014
00015 #define FACTORY_REF(type) __Get ## type
00016
00017 namespace engine {
00018
00019 class Transform;
00020
00021 class Component : public Saveable {
00022 public:
00023   Component();
00024   Component(std::string name);
00025   virtual ~Component();
00026
00027   virtual void begin_play();
00028   virtual void end_play();
00029
00030   virtual void render();
00031   virtual void editor_render();
00032
00033   virtual void tick(float dt);
00034   virtual void init();
00035   virtual void destroy();
00036
00037   virtual void DrawWidget() =0;
00038
00039   Transform* GetOwner();
00040   void SetOwner(Transform* own);
00041
00042   std::string GetName();
00043
00044   virtual json Save() override =0;
00045   virtual bool Load(json value) override =0;
00046   uint64_t GetUID();
00047   std::string name;
00048   std::string component_type;
00049   uint64_t uid;
00050   Transform* owner;
00051
00052   void ComponentDisplay();
00053 };
00054
00055 }
```

9.13 src/engine/editor.cpp File Reference

```
#include "editor.h"
#include "../imgui/imgui.h"
#include "../imgui/imgui_impl_glfw.h"
#include "../imgui/imgui_impl_opengl3.h"
#include "mesh_component.h"
#include "gui_util.h"
#include "opengl_renderer.h"
#include "physics.h"
#include <GL/gl.h>
#include <glm/geometric.hpp>
#include <glm/glm.hpp>
#include <deque>
```

```
#include "camera_component.h"
#include "../imgui/imfilebrowser.h"
#include "../imgui/icon_fonts.h"
```

Namespaces

- namespace engine
- namespace engine::editor

Enumerations

- enum class engine::editor::PrefsPage { engine::editor::APPEARANCE , engine::editor::GENERAL }

Functions

- ImGui::FileBrowser engine::editor::saveBrowser (ImGuiFileBrowserFlags_EnterNewFilename)

Variables

- bool engine::editor::debug_panel_vis = false
- bool engine::editor::asset_panel_vis = false
- bool engine::editor::hierarchy_vis = false
- bool engine::editor::inspector_vis = false
- bool engine::editor::actions_vis = false
- bool engine::editor::preferences_vis = false
- bool engine::editor::scene_load_menu_open = false
- ImGui::FileBrowser engine::editor::loadBrowser
- int engine::editor::fps_smooth_samples = 100
- PrefsPage engine::editor::current_page = PrefsPage::GENERAL

9.14 src/engine/editor.h File Reference

```
#include "../imgui/imgui.h"
#include "engine.h"
```

Classes

- class engine::editor::EditorInstance

Namespaces

- namespace engine
- namespace engine::editor

9.15 editor.h

Go to the documentation of this file.

```
00001 #pragma once
00002
00003 #include "../imgui/imgui.h"
00004 #include "engine.h"
00005 namespace engine::editor {
00006
00007 class EditorInstance {
00008 public:
00009   EditorInstance();
00010   ~EditorInstance();
00011   void Run();
00012   void DebugPanel();
00013   void ActionsPanel();
00014   void AssetPanel();
00015   void Inspector();
00016   void SceneHierarchy();
00017   void MenuBar();
00018
00019
00020   void TransportPanel();
00021
00022
00023   void OpenSceneLoadMenu();
00024   void OpenSceneSaveMenu();
00025
00026   void PreferencesWindow();
00027   void DrawTransformGizmo();
00028   void SceneLoadMenu();
00029
00030
00031   void DrawGrid();
00032   EngineInstance* engine;
00033
00034 private:
00035
00036   bool game_running_switch = false;
00037
00038   EditorCameraComponent* editor_camera = nullptr;
00039   CameraComponent* game_start_camera = nullptr;
00040
00041   unsigned int grid_vao;
00042   unsigned int gizmo_vao;
00043   Transform *hierarchy_selected = nullptr;
00044   void HierarchyLevel(Transform* tr);
00045 };
00046
00047 }
```

9.16 src/engine/engine.cpp File Reference

```
#include "engine.h"
#include "input.h"
#include "opengl_renderer.h"
#include "mesh_component.h"
#include "physics.h"
#include <GLFW/glfw3.h>
#include <GL/glew.h>
#include "../imgui/imgui.h"
#include "../imgui/imgui_impl_glfw.h"
#include "../imgui/imgui_impl_opengl3.h"
#include "sound_manager.h"
```

Namespaces

- namespace engine

Functions

- EngineInstance * engine::Engine ()
- EngineInstance * engine::CreateEngine (EngineCreateInfo &create_info)

Variables

- EngineInstance * engine::instance = nullptr

9.17 src/engine/engine.h File Reference

```
#include <concepts>
#include <vector>
#include "../s7/s7.h"
#include "opengl_renderer.h"
#include "physics.h"
#include "scene.h"
#include "script_engine.h"
#include "scene_loader.h"
#include "asset_manager.h"
#include "input.h"
#include "camera_component.h"
#include "sound_manager.h"
```

Classes

- struct engine::EngineCreateInfo
- class engine::EngineInstance

Namespaces

- namespace engine

Functions

- EngineInstance * engine::CreateEngine (EngineCreateInfo &create_info)
- EngineInstance * engine::Engine ()

9.18 engine.h

Go to the documentation of this file.

```
00001 /*
00002  * * * * * *
00003  *
00004  *     _/        _/_/_/    _/_/_/  _/    _/  _/_/_/_/_/
00005  *    _/          _/    _/        _/    _/       _/
00006  *   _/          _/    _/  _/_/  _/_/_/_/      _/
00007  *  _/          _/    _/    _/  _/    _/      _/
00008  * _/_/_/_/  _/_/_/    _/_/_/  _/    _/      _/
00009  *
00010  *
00011  *    hyper minimalist editor based game engine
00012  *
00013  *
00014  * * * * * *
00015  *
00016  * This engine is proudly AI free, and will remain so
00017  *
00018  * * * * * *
00019  */
00020
00021 #pragma once
00022
00023 #include <concepts>
00024
00025 #include <vector>
00026
00027 #include "../s7/s7.h"
00028
00029 #include "opengl_renderer.h"
00030 #include "physics.h"
00031 #include "scene.h"
00032 #include "script_engine.h"
00033 #include "scene_loader.h"
00034 #include "asset_manager.h"
00035 #include "input.h"
00036 #include "camera_component.h"
00037 #include "sound_manager.h"
00038
00039
00040 namespace engine {
00041
00042 class AssetManager;
00043 class SceneLoader;
00044 class OpenGLRenderer;
00045 class PhysicsHandler;
00046 struct Ray;
00047 struct EngineCreateInfo {
00048   std::string game_name;
00049   uint16_t initial_window_width;
00050   uint16_t initial_window_height;
00051   std::string startup_scene;
00052 };
00053
00054 class EngineInstance {
00055 public:
00056   EngineInstance(EngineCreateInfo& info);
00057   ~EngineInstance();
00058   void Run();
00059   std::string GetVersionString();
00060   SceneLoader* GetSceneLoader();
00061   OpenGLRenderer *GetRenderer();
00062   GLFWwindow *GetWindow();
00063   AssetManager* GetAssetManager();
00064   PhysicsHandler* GetPhysics();
00065   SoundManager* GetSoundManager();
00066
00067   void Stop();
00068
00069
00070
00071   Component *GetLocalPlayer();
00072   void SetLocalPlayer(Component* local_player);
00073
00074
00075
00076
00077
00078   void Initialize();
00079
00080   std::vector<std::string> GetRegisteredComponentsList();
00081
00082   Component *CreateComponent(std::string component_type_name);
```

```
00083   Component* LoadComponent(std::string component_type_name,json value);
00084   void RegisterComponent(std::string type_name, Component* (*factory)(void))
00085   {
00086       registered_component_types[type_name] = factory;
00087   };
00088
00089
00090
00091   InputHandler* GetInput();
00092
00093 private:
00094   std::string startup_scene;
00095   Component* local_player = nullptr;
00096   SoundManager* sound_manager;
00097   void RegisterBuiltinComponents();
00098   glm::vec2 window_size;
00099   GLFWwindow *window;
00100   AssetManager* asset_loader;
00101   SceneLoader* scene_loader;
00102   s7_scheme *scheme_interpreter;
00103   OpenGLRenderer* renderer;
00104   InputHandler* input_handler;
00105   PhysicsHandler* physics_handler;
00106   std::map<std::string, Component* (*) ()> registered_component_types;
00107 };
00108
00109 EngineInstance* CreateEngine(EngineCreateInfo& create_info);
00110
00111 EngineInstance *Engine();
00112 }
```

9.19 src/engine/engine_config.cpp File Reference

```
#include "engine_config.h"
```

Namespaces

- namespace engine

9.20 src/engine/engine_config.h File Reference

```
#include "../json/json.hpp"
#include <glm/glm.hpp>
#include <variant>
```

Classes

- class engine::EngineConfig

Namespaces

- namespace engine

Typedefs

- using engine::ConfigValue = std::variant<int, float, std::string,glm::vec2, glm::vec3, glm::vec4>

9.21 engine_config.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "../json/json.hpp"
00003 #include <glm/glm.hpp>
00004 #include <variant>
00005
00006 using json = nlohmann::json;
00007
00008 namespace engine
00009 {
00010
00011 using ConfigValue = std::variant<int, float, std::string,glm::vec2, glm::vec3, glm::vec4>;
00012
00013 class EngineConfig
00014 {
00015 public:
00016   EngineConfig();
00017   EngineConfig(json value);
00018   ~EngineConfig();
00019   std::map<std::string,ConfigValue> current_config;
00020
00021   ConfigValue GetConfigValue(std::string val);
00022
00023
00024 };
00025
00026 }
```

9.22 src/engine/gizmo.h File Reference

Classes

- class engine::editor::Gizmo

Namespaces

- namespace engine
- namespace engine::editor

9.23 gizmo.h

Go to the documentation of this file.

```
00001 #pragma once
00002
00003 namespace engine::editor {
00004
00005 class Gizmo {
00006 public:
00007   Gizmo();
00008   virtual void draw()=0;
00009 protected:
00010   unsigned int vao;
00011   unsigned int vbo;
00012 };
00013
00014 }
```

9.24 src/engine/gl_util.h File Reference

```
#include <glm/glm.hpp>
```

Classes

- struct BaseUniforms

9.25 gl_util.h

Go to the documentation of this file.

```
00001 #pragma once
00002
00003 #include <glm/glm.hpp>
00004
00005
00006
00007 //these are the uniforms that are to be implemented by all
00008 //shaders, as such, they need to be present in every draw call
00009 struct BaseUniforms {
00010   glm::mat4 model;
00011   glm::mat4 camera;
00012 };
```

9.26 src/engine/gui_util.cpp File Reference

```
#include "gui_util.h"
#include "../imgui/imgui.h"
```

Namespaces

- namespace engine
- namespace engine::gui

Functions

- void engine::gui::TextCentered (const char *text)

9.27 src/engine/gui_util.h File Reference

Namespaces

- namespace engine
- namespace engine::gui

Functions

- void engine::gui::TextCentered (const char *text)

9.28 gui_util.h

Go to the documentation of this file.

```
00001 #pragma once
00002
00003 namespace engine::gui {
00004
00005 void TextCentered(const char *text);
00006
00007 }
```

9.29 src/engine/input.cpp File Reference

```
#include "engine.h"
#include <GLFW/glfw3.h>
#include "input.h"
```

Namespaces

- namespace engine

Functions

- void engine::InputHandlerKeyCallback (GLFWwindow *window, int key, int scancode, int action, int mods)
- void engine::InputHandlerMousePositionCallback (GLFWwindow *window, double xpos, double ypos)
- void engine::InputHandlerMouseButtonCallback (GLFWwindow *window, int button, int action, int mods)
- int engine::MakeKeyval (int key, int mod)

9.30 src/engine/input.h File Reference

```
#include <GLFW/glfw3.h>
#include <map>
#include <glm/glm.hpp>
```

Classes

- class engine::InputHandler

Namespaces

- namespace engine

Enumerations

- enum class engine::KeyState { engine::KEY_UP , engine::KEY_PRESSED , engine::KEY_DOWN , engine::KEY_RELEASED }

Functions

- void engine::InputHandlerKeyCallback (GLFWwindow ∗window, int key, int scancode, int action, int mods)
- void engine::InputHandlerMousePositionCallback (GLFWwindow ∗window, double xpos, double ypos)
- void engine::InputHandlerMouseButtonCallback (GLFWwindow ∗window, int button, int action, int mods)

9.31 input.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include <GLFW/glfw3.h>
00003 #include <map>
00004 #include <glm/glm.hpp>
00005 namespace engine {
00006
00007 enum class KeyState {
00008   KEY_UP,
00009   KEY_PRESSED,
00010   KEY_DOWN,
00011   KEY_RELEASED
00012 };
00013
00014 void InputHandlerKeyCallback(GLFWwindow *window, int key, int scancode,
00015                              int action, int mods);
00016
00017
00018 void InputHandlerMousePositionCallback(GLFWwindow* window, double xpos, double ypos);
00019
00020 void InputHandlerMouseButtonCallback(GLFWwindow* window, int key, int action, int mods);
00021
00022
00023
00024
00025 class InputHandler {
00026 public:
00027   bool KeyDown(int key);
00028   bool KeyDown(int key, int mods);
00029   bool KeyUp(int key);
00030   bool KeyUp(int key, int mods);
00031   bool KeyPressed(int key);
00032   bool KeyPressed(int key, int mods);
00033   bool KeyReleased(int key);
00034   bool KeyReleased(int key, int mods);
00035
00036
00037   void DisableCursor();
00038   void EnableCursor();
00039
00040   bool MouseButtonDown(int button);
00041   bool MouseButtonPressed(int button);
00042   bool MouseButtonReleased(int button);
00043
00044   glm::vec2 MouseDelta();
00045   glm::vec2 MousePosition();
00046
00047   void Update();
00048   void SendInput(int key, int action, int mods);
00049   void SendMousePosition(glm::vec2 position);
00050   void SendMouseButton(int button, int action, int mods);
00051
00052
00053 private:
00054   glm::vec2 mouse_delta_accum = glm::vec2(0.0,0.0);
00055   glm::vec2 mouse_delta = glm::vec2(0.0,0.0);
00056   glm::vec2 last_mouse_pos = glm::vec2(0.0,0.0);
00057
00058   std::map<int, KeyState> key_state;
00059   KeyState leftClick;
00060   KeyState rightClick;
00061
00062
00063
00064
00065 };
00066
00067 }
```

9.32 src/engine/json_util.h File Reference

9.33 json_util.h

Go to the documentation of this file.

```
00001 #pragma once
```

9.34 src/engine/material.cpp File Reference

```
#include "material.h"
#include "../json/json.hpp"
#include "shader.h"
#include <fstream>
#include <iostream>
#include "engine.h"
```

Namespaces

- namespace engine

9.35 src/engine/material.h File Reference

```
#include "texture.h"
#include "shader.h"
#include "asset.h"
#include "gl_util.h"
```

Classes

- class engine::Material

Namespaces

- namespace engine

9.36 material.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "texture.h"
00003 #include "shader.h"
00004 #include "asset.h"
00005 #include "gl_util.h"
00006
00007 namespace engine {
00008
00009   class Shader;
00010   class Asset;
00011
00012     //we hold the uniform sets for each given material in memory
00013     //and they are automatically uploaded, so to change a uniform,
00014     //change the uniform list on the material and it will automatically
00015     //propagate to the gpu
00016     class Material : public Asset
00017     {
00018     public:
00019         Material(std::string path);
00020         bool Reload();
00021         bool Bind(BaseUniforms base);
00022         void DrawEditWidget();
00023       private:
00024         std::string texture_path;
00025         std::string shader_path;
00026         Shader* shader;
00027         Texture* texture;
00028         UniformList uniform_set;
00029     };
00030 };
```

9.37 src/engine/mesh.cpp File Reference

```
#include "mesh.h"
#include <GL/glew.h>
```

Namespaces

- namespace engine

9.38 src/engine/mesh.h File Reference

```
#include <vector>
#include <stdint.h>
#include <glm/glm.hpp>
#include <GL/glew.h>
#include "../dflib/dflib.h"
#include "gl_util.h"
#include "material.h"
```

Classes

- struct Vertex
- class engine::Mesh

Namespaces

- namespace engine

9.39 mesh.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include <vector>
00003 #include <stdint.h>
00004 #include <glm/glm.hpp>
00005
00006 #include <GL/glew.h>
00007 #include "../dflib/dflib.h"
00008 #include "gl_util.h"
00009 #include "material.h"
00010
00011 class Material;
00012
00013 struct Vertex {
00014   glm::vec3 position;
00015   glm::vec3 normal;
00016   glm::vec2 t_coord;
00017 };
00018
00019 namespace engine {
00020
00021 //the mesh class represents an instance of the given
00022 //data on both the CPU and GPU, and the lifetime of
00023 //the data on the GPU depends entirely on the lifetime
00024 //of this object.  Therefore, to load and deload meshes,
00025 //you `new` and `delete` instances of this class
00026
00027 //meshes are grouped into models, so the actual handling
00028 //of that mesh memory is done in `model.cpp`
00029 class Mesh : public Asset{
00030 public:
00031   Mesh();
00032   ~Mesh();
00033   void SetVertices(std::vector<Vertex> v);
00034   void SetIndices(std::vector<unsigned int> v);
00035   bool Rebuffer();
00036   bool Draw(BaseUniforms base, Material* material);
00037   bool DrawWireframe(BaseUniforms base, Material* material);
00038 private:
00039   std::vector<Vertex> vertices;
00040   std::vector<unsigned int> indices;
00041   GLuint vertex_array_handle;
00042   GLuint vertex_buffer_handle;
00043   GLuint index_buffer_handle;
00044 };
00045 }
```

9.40 src/engine/mesh_component.cpp File Reference

```
#include "mesh_component.h"
#include "component.h"
#include "engine.h"
#include "../imgui/imgui.h"
#include "../imgui/icon_fonts.h"
```

Namespaces

- namespace engine

Functions

- engine::FACTORY (MeshComponent)

9.41 src/engine/mesh_component.h File Reference

```
#include "component.h"
#include "mesh.h"
#include "material.h"
```

Classes

- class engine::MeshComponent

Namespaces

- namespace engine

Functions

- engine::FACTORY (MeshComponent)

9.42 mesh_component.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "component.h"
00003 #include "mesh.h"
00004 #include "material.h"
00005 namespace engine {
00006
00007 class MeshComponent : public Component {
00008 public:
00009   virtual void render() override;
00010   virtual void init() override;
00011   virtual void destroy() override;
00012   bool SetMesh(std::string mesh);
00013   bool SetMaterial(std::string material);
00014   void DrawWidget() override;
00015   MeshComponent(std::string name);
00016
00017   MeshComponent(json value);
00018
00019   bool Load(json value) override;
00020   json Save() override;
00021 private:
00022   std::string mesh_name;
00023   bool mesh_valid = false;
00024   std::string material_name;
00025   bool material_valid = false;
00026 };
00027
00028
00029 FACTORY(MeshComponent);
00030
00031
00032
00033 }
```

9.43 src/engine/model.cpp File Reference

```
#include "model.h"
#include "mesh.h"
#include <assimp/Importer.hpp>
#include <assimp/scene.h>
#include <assimp/postprocess.h>
#include <iostream>
#include "../dflib/dflib.h"
#include <filesystem>
```

Namespaces

- namespace engine

Variables

- std::map< std::string, Model * > engine::modelCache

9.44 src/engine/model.h File Reference

```
#include <vector>
#include "mesh.h"
#include "material.h"
#include "texture.h"
#include <assimp/scene.h>
#include <map>
#include "asset.h"
```

Classes

- class engine::Model

Namespaces

- namespace engine

Functions

- Model * engine::GetModel (std::string path)

9.45 model.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include <vector>
00003 #include "mesh.h"
00004 #include "material.h"
00005 #include "texture.h"
00006 #include <assimp/scene.h>
00007 #include <map>
00008 #include "asset.h"
00009
00010 namespace engine {
00011 class Mesh;
00012 class Asset;
00013
00014 class Model : public Asset {
00015 public:
00016   Model(std::string path);
00017   ~Model();
00018   std::vector<Mesh*> GetMeshes();
00019 private:
00020   bool ProcessNode(aiNode *node, const aiScene *scene);
00021   Mesh* ProcessMesh(aiMesh *mesh, const aiScene *scene);
00022   std::vector<Mesh*> meshes;
00023 };
00024
00025 Model* GetModel(std::string path);
00026
00027 }
```

9.46 src/engine/opengl_renderer.cpp File Reference

```
#include "opengl_renderer.h"
#include <GLFW/glfw3.h>
#include "../imgui/imgui.h"
#include "../imgui/imgui_impl_glfw.h"
#include "../imgui/imgui_impl_opengl3.h"
#include "../imgui/icon_fonts.h"
#include "input.h"
#include "../stb/stb_image.h"
```

Namespaces

- namespace engine

Macros

- #define STB_IMAGE_IMPLEMENTATION

Functions

- void APIENTRY engine::MessageCallback (GLenum source, GLenum type, GLuint id, GLenum severity, GLsizei length, const GLchar *message, const void *userParam)
- void engine::window_size_callback (GLFWwindow *window, int width, int height)

Variables

- int engine::window_w = 0
- int engine::window_h = 0

9.46.1 Macro Definition Documentation

9.46.1.1 STB_IMAGE_IMPLEMENTATION

```
#define STB_IMAGE_IMPLEMENTATION
```

9.47 src/engine/opengl_renderer.h File Reference

```
#include <GL/glew.h>
#include <GLFW/glfw3.h>
#include "engine.h"
```

Classes

- class engine::OpenGLRenderer

Namespaces

- namespace engine

9.48 opengl_renderer.h

Go to the documentation of this file.

```
00001 #pragma once
00002
00003 #include <GL/glew.h>
00004 #include <GLFW/glfw3.h>
00005 #include "engine.h"
00006
00007 namespace engine {
00008 class OpenGLRenderer {
00009 public:
00010   OpenGLRenderer(GLFWwindow** smuggle, struct EngineCreateInfo& create_info);
00011   ~OpenGLRenderer();
00012   bool ShouldClose();
00013
00014
00015   float DeltaTime();
00016
00017   void StartFrame();
00018
00019
00020   void EndFrame();
00021
00022   void InitForImGui();
00023
00024   int WindowWidth();
00025   int WindowHeight();
00026
00027 private:
00028   int window_width;
00029   int window_height;
00030   double last;
00031   float delta_time;
00032   GLFWwindow* window;
00033 };
00034 }
```

9.49 src/engine/physics.cpp File Reference

```
#include "physics.h"
#include <vector>
#include "../game/collision_component.h"
#include "../game/trigger_component.h"
```

Namespaces

- namespace engine

Functions

- bool engine::TestRayOBBIntersection (glm::vec3 ray_origin, glm::vec3 ray_direction, glm::vec3 aabb_min, glm::vec3 aabb_max, glm::mat4 ModelMatrix, float &intersection_distance)

9.50 src/engine/physics.h File Reference

```
#include <glm/glm.hpp>
#include "camera_component.h"
```

Classes

- struct engine::Ray
- class engine::PhysicsHandler

Namespaces

- namespace engine

Functions

- bool engine::TestRayOBBIntersection (glm::vec3 ray_origin, glm::vec3 ray_direction, glm::vec3 aabb_min, glm::vec3 aabb_max, glm::mat4 ModelMatrix, float &intersection_distance)

9.51 physics.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include <glm/glm.hpp>
00003 #include "camera_component.h"
00004
00005 namespace engine
00006 {
00007
00008 class CollisionComponent;
00009 class TriggerComponent;
00010
00011 struct Ray
00012 {
00013   glm::vec3 position;
00014   glm::vec3 direction;
00015 };
00016
00017
00018
00019 bool TestRayOBBIntersection(
00020     glm::vec3 ray_origin,    // Ray origin, in world space
00021     glm::vec3 ray_direction, // Ray direction (NOT target position!), in world
00022                              // space. Must be normalize()'d.
00023     glm::vec3 aabb_min, // Minimum X,Y,Z coords of the mesh when not transformed
00024                         // at all.
00025     glm::vec3 aabb_max, // Maximum X,Y,Z coords. Often aabb_min*-1 if your mesh
00026                         // is centered, but it's not always the case.
00027     glm::mat4 ModelMatrix, // Transformation applied to the mesh (which will
00028                            // thus be also applied to its bounding box)
00029     float &intersection_distance // Output : distance between ray_origin and the
00030                                  // intersection with the OBB
00031 );
00032
00033 class PhysicsHandler {
00034 public:
00035   bool CheckTrigger(CollisionComponent* collider, int trigger_group);
00036
00037   void RegisterCollider(CollisionComponent* collider);
00038   void DeregisterCollider(CollisionComponent *collider);
00039   std::vector<CollisionComponent*>* GetColliders();
00040
00041   void RegisterTrigger(TriggerComponent* trigger);
00042   void DeregisterTrigger(TriggerComponent* trigger);
00043   std::vector<TriggerComponent*>* GetTriggers();
00044
00045 private:
00046   std::vector<CollisionComponent*> registered_colliders;
00047   std::vector<TriggerComponent*> registered_triggers;
00048 };
00049
00050
00051
00052
00053
00054
00055
00056
00057
00058
00059
00060
00061 }
```

9.52 src/engine/saveable.h File Reference

```
#include "../json/json.hpp"
```

Classes

- class engine::Saveable

Namespaces

- namespace engine

Macros

- #define PROPERTY_SAVE(prop)
- #define PROPERTY_LOAD(prop)
- #define PROPERTY_CHECK(prop)

9.52.1 Macro Definition Documentation

9.52.1.1 PROPERTY_CHECK

```
#define PROPERTY_CHECK(
                prop)
```

Value:

```
if(!value.contains(#prop)) { \
std::cout « "property `" « #prop « "` failed to load on component " « this->name.c_str() «" UID: " «
      this->uid « "\n"; \
  return false;}
```

9.52.1.2 PROPERTY_LOAD

```
#define PROPERTY_LOAD(
                prop)
```

Value:

```
if(value.contains(#prop)) { \
prop = value[#prop]; \
} else { \
std::cout « "property `" « #prop « "` failed to load on component " « this->name.c_str() «" UID: " «
      this->uid « "\n"; \
  return false; }
```

9.52.1.3 PROPERTY_SAVE

```
#define PROPERTY_SAVE(
                prop)
```

Value:

```
value[#prop] = prop;
```

9.53 saveable.h

Go to the documentation of this file.

```
00001 #pragma once
00002
00003
00004 #include "../json/json.hpp"
00005 using json = nlohmann::json;
00006
00007
00008 #define PROPERTY_SAVE(prop) value[#prop] = prop;
00009
00010 #define PROPERTY_LOAD(prop) if(value.contains(#prop)) { \
00011 prop = value[#prop]; \
00012 } else { \
00013 std::cout « "property `" « #prop « "` failed to load on component " « this->name.c_str() «" UID: " «
      this->uid « "\n"; \
00014   return false; }
00015
00016 #define PROPERTY_CHECK(prop) if(!value.contains(#prop)) { \
00017 std::cout « "property `" « #prop « "` failed to load on component " « this->name.c_str() «" UID: " «
      this->uid « "\n"; \
00018   return false;}
00019
00020
00021
00022 namespace engine {
00023 class Saveable {
00024   virtual json Save() =0;
00025   virtual bool Load(json value)=0;
00026 };
00027 }
```

9.54 src/engine/scene.cpp File Reference

```
#include "scene.h"
#include "camera_component.h"
#include "engine.h"
```

Namespaces

- namespace engine

9.55 src/engine/scene.h File Reference

```
#include "transform.h"
#include "camera_component.h"
#include "saveable.h"
```

Classes

- class engine::Scene

Namespaces

- namespace engine

9.56 scene.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "transform.h"
00003 #include "camera_component.h"
00004 #include "saveable.h"
00005 namespace engine
00006 {
00007
00008 class CameraComponent;
00009
00010
00011
00012 class Scene : public Saveable {
00013 public:
00014   Scene();
00015   Scene(std::string name);
00016   ~Scene();
00017   void Render();
00018   void Tick(float dt);
00019   void EditorRender();
00020   json Save() override;
00021   bool Load(json value) override;
00022
00023   Transform *GetRoot();
00024   bool IsValid();
00025
00026   void SetCamera(CameraComponent* cam);
00027   void SetMainCamera(CameraComponent* cam);
00028   void SetCameraToMain();
00029
00030   Scene *Clone();
00031
00032   glm::mat4 GetCurrentCameraMatrix();
00033   glm::vec3 GetCurrentCameraPosition();
00034   CameraComponent* GetCurrentCamera();
00035 private:
00036   CameraComponent* current_camera;
00037   CameraComponent* main_camera;
00038   std::string name;
00039   std::string path;
00040   Transform* root;
00041   bool finalized;
00042 };
00043
00044 }
```

9.57 src/engine/scene_loader.cpp File Reference

```
#include "scene_loader.h"
#include "camera_component.h"
#include "mesh_component.h"
#include "engine.h"
#include <fstream>
```

Namespaces

- namespace engine

9.58 src/engine/scene_loader.h File Reference

```
#include "scene.h"
#include "../dflib/dflib.h"
#include <future>
```

Classes

- class engine::SceneLoader

Namespaces

- namespace engine

9.59 scene_loader.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "scene.h"
00003 #include "../dflib/dflib.h"
00004 #include <future>
00005
00006 namespace engine {
00007
00008 class Scene;
00009
00010
00011 class SceneLoader {
00012 public:
00013   SceneLoader();
00014   ~SceneLoader();
00015   //std::future<Scene*> CueScene(std::string path); // async load scene from json dump to on deck
      scene, for streaming or quickswapping scenes,
00016   bool LoadScene(std::string path, void* data);
00017
00018   void LoadDefaultScene();                 // immediately load the startup scene
00019   void SaveScene(std::string path);
00020   void DeloadScene();
00021
00022   bool UpdateCurrentScene(float dt);
00023   bool RenderCurrentScene();
00024   Scene* GetCurrentScene();
00025
00026   void SaveAsCheckpoint();
00027   void RewindToCheckpoint();
00028
00029   void* GetPersistentData();
00030
00031
00032
00033 private:
00034   void* persistant_data = nullptr;
00035   Scene *current_scene = nullptr;
00036   Scene *rewind_scene = nullptr;
00037   Scene* scene_in_progress_load = nullptr;
00038   bool LoadScene(std::string path); // sync load scene from json dump to current scene
00039 };
00040 }
```

9.60 src/engine/screen_shader.cpp File Reference

```
#include "screen_shader.h"
```

Namespaces

- namespace engine

Variables

- float engine::quad_data [ ]

9.61 src/engine/screen_shader.h File Reference

```
#include "shader.h"
```

Classes

- class engine::ScreenShader

Namespaces

- namespace engine

9.62 screen_shader.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "shader.h"
00003 namespace engine {
00004 class ScreenShader : public Shader {
00005
00006
00007   ScreenShader(std::string path);
00008   ~ScreenShader();
00009   void Apply();
00010 };
00011
00012 }
```

9.63 src/engine/script_component.cpp File Reference

```
#include "script_component.h"
#include "../s7/s7.h"
```

Namespaces

- namespace engine

Functions

- void engine::tick (float dt)
- void engine::init ()

9.64 src/engine/script_component.h File Reference

```
#include "component.h"
#include <string>
#include <variant>
#include "script_engine.h"
#include "../s7/s7.h"
```

Classes

- class engine::ScriptComponent

Namespaces

- namespace engine

9.65 script_component.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "component.h"
00003 #include <string>
00004 #include <variant>
00005 #include "script_engine.h"
00006 #include "../s7/s7.h"
00007 namespace engine {
00008
00009 class ScriptComponent : public Component {
00010 public:
00011   virtual void tick(float dt) override;
00012   virtual void init() override;
00013   virtual void destroy() override;
00014   void ReloadScripts();
00015 private:
00016   ScriptData init_script;
00017   ScriptData tick_script;
00018   ScriptData destroy_script;
00019 };
00020
00021 }
```

9.66 src/engine/script_engine.h File Reference

```
#include "../s7/s7.h"
#include "asset_manager.h"
#include <variant>
#include <string>
```

Classes

- struct engine::ScriptLiteral
- struct engine::ScriptFile
- class engine::ScriptEngine
- class engine::SchemeScriptEngine

Namespaces

- namespace engine

Typedefs

- typedef std::variant< ScriptLiteral, ScriptFile > engine::ScriptData

9.67 script_engine.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "../s7/s7.h"
00003
00004 #include "asset_manager.h"
00005
00006 #include <variant>
00007 #include <string>
00008
00009 namespace engine {
00010
00011 struct ScriptLiteral {
00012   std::string script;
00013 };
00014
00015 struct ScriptFile {
00016   std::string script_path;
00017   std::string cached_script;
00018   bool stale;
00019 };
00020
00021 typedef std::variant<ScriptLiteral, ScriptFile> ScriptData;
00022
00023 class ScriptEngine {
00024 public:
00025   virtual void eval_string() = 0;
00026 };
00027
00028 class SchemeScriptEngine : public ScriptEngine {
00029
00030 public:
00031    SchemeScriptEngine();
00032    ~SchemeScriptEngine();
00033    void eval_string() override;
00034 private:
00035   s7_scheme* scheme_handle;
00036 };
00037
00038 }
```

9.68 src/engine/shader.cpp File Reference

```
#include "shader.h"
#include "../dflib/dflib.h"
#include <glm/ext.hpp>
#include <variant>
#include <fstream>
#include "../json/json.hpp"
#include "opengl_renderer.h"
```

Namespaces

- namespace engine

9.69 src/engine/shader.h File Reference

```
#include <vector>
#include <variant>
#include <string>
#include <glm/glm.hpp>
#include <GL/glew.h>
#include "../dflib/dflib.h"
#include <map>
#include "asset.h"
```

Classes

- struct engine::Uniform_i
- struct engine::Uniform_ui
- struct engine::Uniform_f
- struct engine::Uniform_d
- struct engine::Uniform_mat4
- struct engine::Uniform_vec3
- class engine::UniformList
- class engine::Shader

Namespaces

- namespace engine

Typedefs

- using engine::Uniform

9.70 shader.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include <vector>
00003 #include <variant>
00004 #include <string>
00005 #include <glm/glm.hpp>
00006 #include <GL/glew.h>
00007 #include "../dflib/dflib.h"
00008 #include <map>
00009 #include "asset.h"
00010
00011 namespace engine {
00012
00013 class Asset;
00014
00015 struct Uniform_i {
00016   int value;
00017   Uniform_i(int v)
00018   {
00019     value = v;
00020   }
00021 };
00022 struct Uniform_ui {
00023   unsigned int value;
00024 };
00025 struct Uniform_f {
```

```
00026   float value;
00027   Uniform_f(float f)
00028   {
00029     value = f;
00030   }
00031 };
00032 struct Uniform_d {
00033   double value;
00034 };
00035 struct Uniform_mat4 {
00036   glm::mat4 value;
00037   Uniform_mat4(glm::mat4 m) {
00038     value = m;
00039   }
00040 };
00041
00042 struct Uniform_vec3 {
00043   glm::vec3 value;
00044   Uniform_vec3(glm::vec3 v) {
00045     value = v;
00046   }
00047 };
00048
00049 using Uniform =
00050     std::variant<Uniform_vec3,Uniform_i, Uniform_ui, Uniform_f, Uniform_d, Uniform_mat4>;
00051
00052 class UniformList {
00053 public:
00054   std::map<std::string,Uniform> uniforms;
00055 };
00056
00057
00058 class Shader : public Asset{
00059 public:
00060   Shader(std::string path);
00061   ~Shader();
00062   bool SetUniforms(UniformList& u);
00063   bool SetUniform(std::string key, Uniform u);
00064   void Reload();
00065   bool Bind();
00066 protected:
00067   unsigned int shader_handle = 0;
00068   std::string vertex_src_str;
00069   std::string fragment_src_str;
00070 };
00071
00072 };
```

9.71 src/engine/sound.cpp File Reference

```
#include "sound.h"
#include "engine.h"
```

Namespaces

- namespace engine

9.72 src/engine/sound.h File Reference

```
#include "asset.h"
#include "../miniaudio/miniaudio.h"
```

Classes

- class engine::Sound

Namespaces

- namespace engine

9.73 sound.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "asset.h"
00003 #include "../miniaudio/miniaudio.h"
00004
00005
00006 namespace engine
00007 {
00008
00009 class Sound : public Asset {
00010 public:
00011   Sound(std::string path);
00012   void PlayLooped();
00013
00014 private:
00015     ma_sound* sound_handle;
00016 };
00017
00018
00019
00020
00021 }
```

9.74 src/engine/sound_manager.cpp File Reference

```
#include "sound_manager.h"
#include <iostream>
```

Namespaces

- namespace engine

9.75 src/engine/sound_manager.h File Reference

```
#include "../miniaudio/miniaudio.h"
#include <string>
```

Classes

- class engine::SoundManager

Namespaces

- namespace engine

9.76 sound_manager.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "../miniaudio/miniaudio.h"
00003 #include <string>
00004
00005
00006 namespace engine {
00007 class SoundManager {
00008 public:
00009   SoundManager();
00010   ~SoundManager();
00011   void PlaySound(std::string sound);
00012   ma_sound* LoadSound(std::string path);
00013 private:
00014     ma_engine* engine;
00015 };
00016
00017 }
```

9.77 src/engine/texture.cpp File Reference

```
#include "texture.h"
#include "../stb/stb_image.h"
#include "GL/glew.h"
#include <GL/gl.h>
#include <iostream>
```

Namespaces

- namespace engine

9.78 src/engine/texture.h File Reference

```
#include <string>
#include "asset.h"
```

Classes

- class engine::Texture

Namespaces

- namespace engine

9.79 texture.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include <string>
00003 #include "asset.h"
00004 namespace engine {
00005
00006 class Asset;
00007
00008 class Texture : public Asset{
00009 public:
00010   Texture(std::string path);
00011   ~Texture();
00012   void Bind();
00013
00014   unsigned int GetTextureHandle();
00015
00016 private:
00017   unsigned int texture_handle;
00018 };
00019 }
```

9.80 src/engine/theme.h File Reference

9.81 theme.h

Go to the documentation of this file.

```
00001 #ifndef THEME_H_
00002 #define THEME_H_
00003
00004
00005
00006 #endif // THEME_H_
```

9.82 src/engine/transform.cpp File Reference

```
#include "transform.h"
#include <cstdint>
#include <glm/gtc/quaternion.hpp>
#include "util.h"
#include "../imgui/icon_fonts.h"
#include "../imgui/imgui.h"
#include "engine.h"
```

Namespaces

- namespace engine

9.83 src/engine/transform.h File Reference

```
#include "component.h"
#include "saveable.h"
#include <glm/glm.hpp>
#include <glm/ext.hpp>
#include <vector>
#include <string>
#include "../dflib/dflib.h"
```

Classes

- class engine::Transform

Namespaces

- namespace engine

9.84 transform.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include "component.h"
00003 #include "saveable.h"
00004 #include <glm/glm.hpp>
00005 #include <glm/ext.hpp>
00006 #include <vector>
00007 #include <string>
00008 #include "../dflib/dflib.h"
00009
00010 namespace engine {
00011
00012 class Component;
00013
00014 class Transform : public Saveable {
00015 public:
00019   Transform();
00020
00028   Transform(json value);
00029
00035   Transform(std::string name);
00036
00041   ~Transform();
00042
00048   void SetName(std::string name);
00049
00055   json Save() override;
00056
00062   bool Load(json value) override;
00063
00067   void ProcessEditorRender();
00068
00072   void ProcessRender();
00073
00077   void ProcessTick(float dt);
00078
00083   void ProcessInit();
00084
00090   void AddComponent(Component* component);
00091
00104   Component* GetComponent(std::string name);
00105
00115   void AddChild(Transform *tr);
00116
00126   Transform *FindTransformByUID(uint64_t id);
00127
00137   Component* FindComponentByUID(uint64_t id);
00138
00150   Transform *FindChildByName(std::string name);
00151
00163   Transform* RemoveChild(Transform *tr);
00164
00165
00166
00172   std::vector<Transform*> GetChildren();
00173
00179   void Translate(glm::vec3 offset);
00180
00181
00187   void Scale(glm::vec3 amount);
00188
00195   void Rotate(glm::vec3 axis, float amount);
00196
00202   glm::vec3 Forward();
00203
00209   glm::vec3 Backward();
```

```
00210
00216   glm::vec3 Up();
00217
00223   glm::vec3 Down();
00224
00230   glm::vec3 Left();
00231
00237   glm::vec3 Right();
00238
00239
00245   glm::mat4 GetModelMatrix();
00246
00254   glm::mat4 GetModelMatrixUnscaled();
00255
00262   glm::vec3 GetGlobalPosition();
00263
00269   glm::vec3 GetLocalPosition();
00270
00276   glm::quat GetGlobalRotation();
00277
00283   glm::quat GetLocalRotation();
00284
00285
00286
00292   glm::vec3 GetGlobalScale();
00293
00299   glm::vec3 GetLocalScale();
00300
00301
00302
00303
00309   std::string GetName();
00310
00316   uint64_t GetUID();
00317
00318
00322   void DrawWidget();
00323
00324   glm::vec3 position;
00325   glm::quat rotation;
00326   glm::vec3 scale;
00327   std::string name;
00328   uint64_t uid;
00329   glm::vec3 rotationEuler; //slaved to the quaternion
00330
00331   Transform *parent = nullptr;
00332   std::vector<Transform *> children;
00333   std::vector<Component*> components;
00334
00335 private:
00336 };
00337
00338 }
```

9.85 src/engine/util.cpp File Reference

```
#include "util.h"
```

Functions

- uint64_t GenerateUID ()
- void SetUIDOffset (uint64_t offset)
- glm::vec3 MoveTowards (glm::vec3 current, glm::vec3 target, float delta)

Variables

- uint64_t uid_ctr = 0

9.85.1 Function Documentation

9.85.1.1 GenerateUID()

```
uint64_t GenerateUID ()
```

9.85.1.2 MoveTowards()

```
glm::vec3 MoveTowards (
            glm::vec3 current,
            glm::vec3 target,
            float delta)
```

9.85.1.3 SetUIDOffset()

```
void SetUIDOffset (
            uint64_t offset)
```

9.85.2 Variable Documentation

9.85.2.1 uid_ctr

```
uint64_t uid_ctr = 0
```

9.86 src/engine/util.h File Reference

```
#include <stdint.h>
#include <glm/glm.hpp>
```

Functions

- uint64_t GenerateUID ()
- void SetUIDOffset (uint64_t offset)
- glm::vec3 MoveTowards (glm::vec3 current, glm::vec3 target, float delta)

9.86.1 Function Documentation

9.86.1.1 GenerateUID()

```
uint64_t GenerateUID ()
```

9.86.1.2 MoveTowards()

```
glm::vec3 MoveTowards (
            glm::vec3 current,
            glm::vec3 target,
            float delta)
```

9.86.1.3 SetUIDOffset()

```
void SetUIDOffset (
            uint64_t offset)
```

9.87 util.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include <stdint.h>
00003 #include <glm/glm.hpp>
00004
00005 //simple incremental uid, there could (theoretically) be collisions if it rolls over
00006 uint64_t GenerateUID();
00007
00008 //should only be called after a fresh wipe and load, and should be set to the highest UID
00009 //in the loaded scene, to prevent collisions with loaded objects
00010 void SetUIDOffset(uint64_t offset);
00011
00012 glm::vec3 MoveTowards(glm::vec3 current, glm::vec3 target, float delta);
```

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